

WHILE WE'RE HERE

SMALL ESSAYS ON ATTENTION, TIME, AND BEING ALIVE

ADAM HICKEY

WHILE WE'RE HERE

SOURCES

smartphone checks – Andrews S et al. PLOS ONE 10(10): e0139004, 2015

starling topological distance – Ballerini M et al. PNAS 105(4): 1232–1237, 2008

physarum tokyo rail – Tero A et al. Science 327(5964): 439–442, 2010

human and bacterial cell counts – Sender R et al. PLOS Biology 14(8): e1002533, 2016

termite nest self organization – Heyde A et al. PNAS 118(5): e2006985118, 2021

honeycomb hexagon mechanism – Nazzi F. Scientific Reports 6: 28341, 2016

working memory four chunks – Cowan N. Behavioral and Brain Sciences 24(1): 87–114, 2001

london taxi hippocampus – Maguire EA et al. PNAS 97(8): 4398–4403, 2000

google effects on memory – Sparrow B et al. Science 333(6043): 776–778, 2011

spider web sensory extension – Japyassú HF & Laland KN. Animal Cognition 20: 375–395, 2017

intel 4004 – Intel Corporation. Intel company timeline, 1971

transformer paper 2017 – Vaswani A et al. Advances in Neural Information Processing Systems 30, 2017

alphafold database scale – EMBL-EBI and Google DeepMind. Public database, 2024

us internet and broadband adoption – Pew Research Center, 2025

us smartphone ownership – Pew Research Center, 2025

first sms and character limit – Vodafone newsroom, 2017

shikoku henro – Japan National Tourism Organization. japan.travel, 2024

camino frances and compostela – Turespaña. spain.info, official tourism portal, 2024

unesco santiago routes france – UNESCO World Heritage Centre. World Heritage List, ref. 868, 1998

sunlight travel time – NASA Space Place, 2024

earth rotation and orbital speed – NASA Science, 2024

resting respiration rate – StatPearls. NCBI Bookshelf, 2023

komodo directional chemoreception – Smithsonian's National Zoo and Conservation Biology Institute. Smithsonian National Zoo animal record, 2024

queen monarch mullerian mimicry – Ritland DB. Ecology 75(3): 732–746, 1994

inferred ad categories – Meta Platforms. Download Your Information, personal export, 2026

advertisers holding your data – Meta Platforms. Download Your Information, personal export, 2026

cities checked into – Meta Platforms. Download Your Information, personal export, 2026

habituation – Rankin CH et al. Neurobiology of Learning and Memory 92(2): 135–138, 2009

© 2026 Adam Hickey. All rights reserved.

Photographs, drawings and diagrams by Adam Hickey except as listed.

starling flock – Adapted from a photograph by Yuri Elizegi, with the bare-tree foreground from a photograph by Vera Emilie A Bomstad, both via Pexels. Composited and color-graded for this book · Pexels License

physarum network – Rob Cruickshank, “Slime mould (P. polycephalum)”, via Wikimedia Commons. Cropped, masked, composited and color-graded for this book · CC BY 2.0 · creativecommons.org/licenses/by/2.0/

oxidized metal – Composited from Poly Haven's “Rusty Metal 04” PBR maps and a photograph of weathered copper by Lana on Unsplash, relit and rebalanced for this book · CC0 · Unsplash License

closing hands – Adapted from a photograph by Tima Miroshnichenko via Pexels. Color-graded and composited for this book · Pexels License

closing cursor – Adapted from a photograph by Artem Podrez via Pexels. Color-graded and composited for this book · Pexels License



While We're Here

Small essays on attention, time, and being alive

ADAM HICKEY

**For my parents,
who taught me to look, and for
Fabiola, who looks with me**

These are small essays about being alive, written mostly on foot.

None of them argue for anything. I have gotten steadily less interested in being right and steadily more interested in noticing things, which is a poor strategy for winning discussions and a good one for most of the rest of it.

Mostly I wanted to know one thing. How much is going on in a familiar room, or on a walk I have taken a thousand times, that I have simply stopped being able to see?

You can read this from the beginning. You can also open it anywhere, read four pages, and put it down. It was built to survive both.

PART I

Look Again

Most of Life Is a Tuesday	011
The Secret Life of Attention	025

PART II

What Are We?

The Beauty of Systems Nobody Designed	043
The Intelligence Outside Your Head	055

PART III

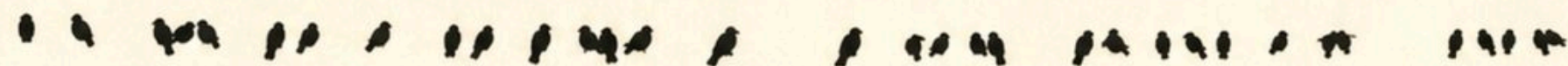
The World Is Changing

The Strange Privilege	071
The Last People Who Remember Waiting	087

PART IV

While We’re Here

The Body Cannot Skip the Hill	103
While We’re Here	117



Am ze pt. e la h. yin g q .d. try lie n h . q . d. l. o. r. e



OBSERVE THE SURFACE

Part I

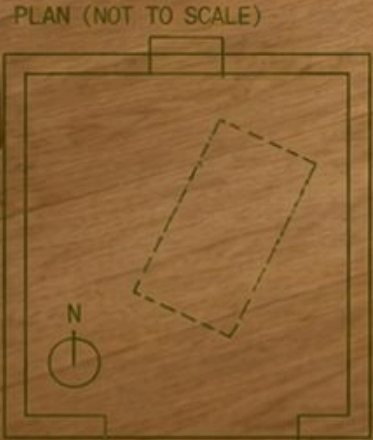
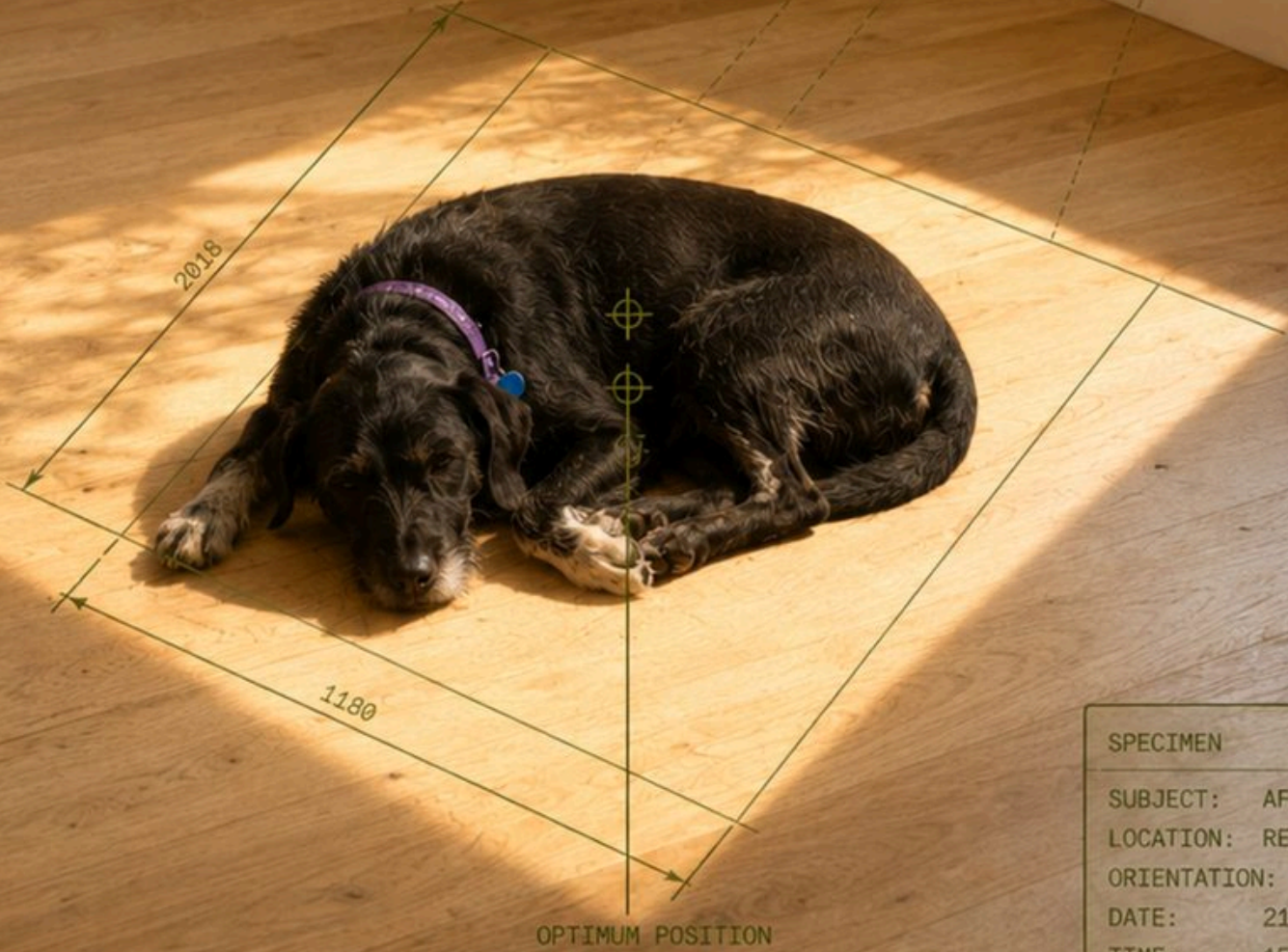
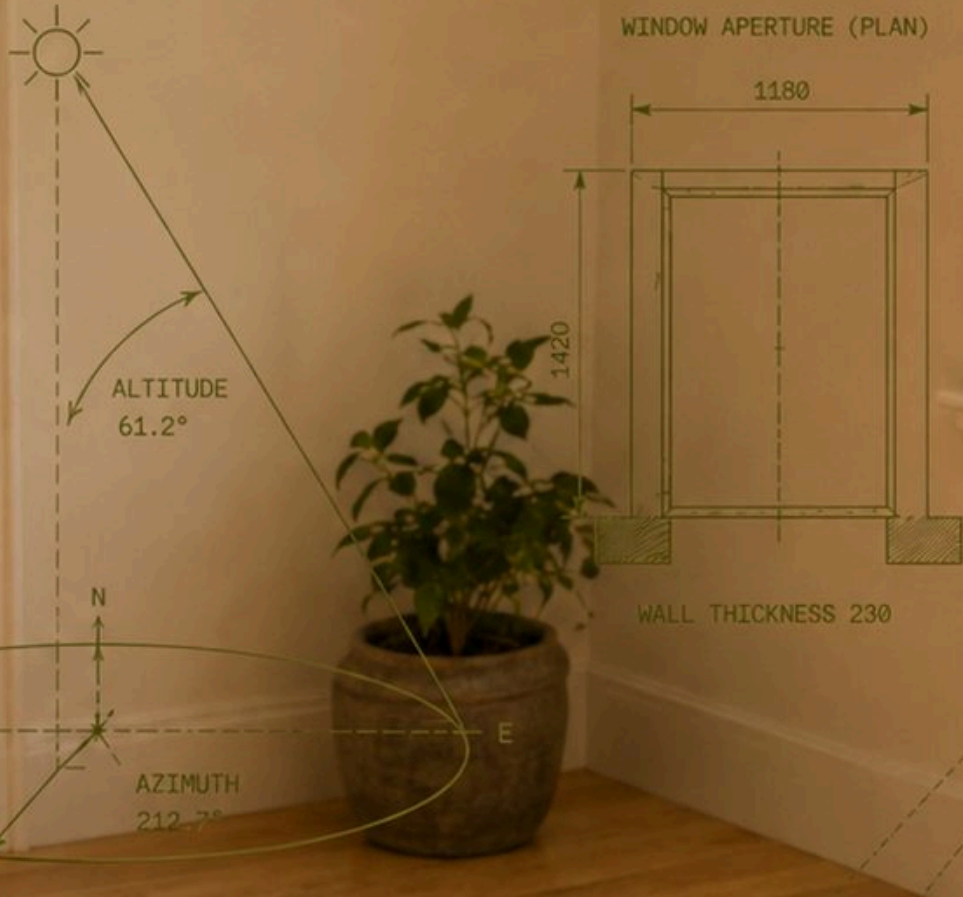
Look Again

Nothing in this part is unfamiliar.
That is the whole of the difficulty.

Attention, beauty, and the case for the ordinary day.

- 01 Most of Life Is a Tuesday
- 02 The Secret Life of Attention

SOLAR GEOMETRY
DATE: 21 / 06
TIME: 15:00
LAT: 45.4167° N
LON: 75.7000° W



SPECIMEN	
SUBJECT:	AFTERNOON LIGHT PATCH
LOCATION:	RESIDENTIAL ROOM
ORIENTATION:	1180
DATE:	21 / 06
TIME:	15:00
SCALE:	1:20
UNITS:	MILLIMETRES

Most of Life Is a Tuesday

Almost everything happens between the moments we think are important.

The Life Between the Milestones

The dog moved twice this afternoon and never woke. The sun came in low and crossed half the floor between two and five, and the dog crossed with it — a slow pursuit conducted entirely in sleep. I measured it once, out of curiosity. About a meter an hour. That is the honest record of the day, and I have started to suspect that days like this one are not the packaging around my life. They are the substance.

Most of life does not happen during the big moments.
It happens in between them.

Not at the wedding, but on a Tuesday night years later
when you're both tired and deciding what to eat.
Not on the vacation, but while making
coffee before work.
Not at the graduation, but during the thousands of
ordinary mornings that follow.
Most of life is quieter than that.

A walk.
A meal.
A familiar room.

And yet we have a strange habit of treating these moments as filler, as though the real life is somewhere else.

We organize our memories around events.
Birthdays. Trips. Weddings. New jobs.
Moves. Achievements.
These become the chapter headings.

But chapter headings are not the book.
The book is everything that happens between them.

If you live for eighty years, only a tiny fraction of that time will contain anything you would call extraordinary.

Most mornings will be ordinary mornings.
Most dinners will not be memorable.
Most conversations will disappear.

If happiness requires extraordinary circumstances, then happiness is necessarily rare. But if ordinary experience can become meaningful, almost your entire life becomes available.

Most of life is not a highlight reel.

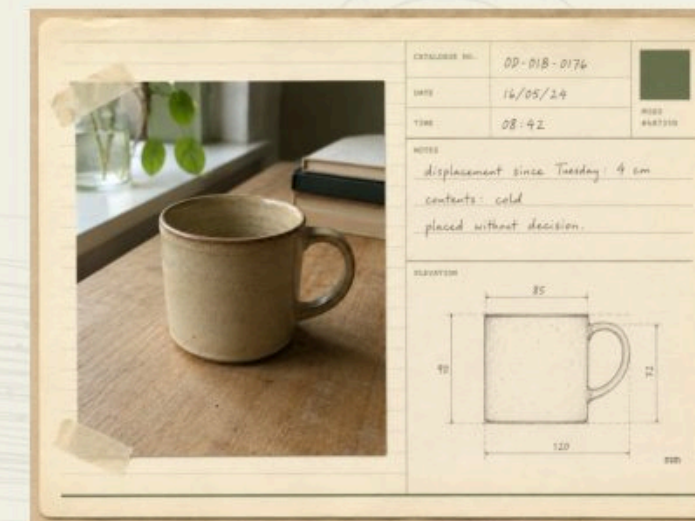
This may be one of the central misunderstandings of adulthood.

We are constantly encouraged to imagine happiness as an event. Something happens. Then you feel different.

You get the house.
Reach the goal.
Take the trip.
Retire.
Finally have enough time.

The future contains the life you're trying to reach. And the present becomes transportation, a bridge between where you are and where you're supposed to be.

The danger is that you can spend decades crossing bridges. Because every destination eventually becomes another starting point. You arrive. The



mind adjusts. And almost immediately, life begins pointing somewhere else.

Children seem to understand something adults gradually forget.

A child can spend twenty minutes watching water move through a gutter.
A stick becomes an object worth carrying home.
Snow is an event.
A dog walking past deserves full attention.

The world hasn't yet been divided so aggressively into important and unimportant things.

Adults become efficient. We've seen ^{the path} the tree before. We know what rain is. Familiarity allows us to move faster.

But familiarity also makes reality disappear. The more certain we become that we know what something is, the less closely we look at it.



same path, same water — 17 May

Attention Changes the World

Travel can make us feel unusually alive for exactly this reason.
Everything becomes visible again.

- Street signs.
- Breakfast.
- People walking to work.
- The way grocery stores are arranged.

You suddenly notice ordinary things because they are unfamiliar.

We usually assume the place created the feeling. Maybe partly.
But another possibility is that attention created the feeling.
The world became interesting because you stopped moving
through it automatically.

Which raises an uncomfortable question: how much of everyday life are
we simply failing to notice?



Brains economize. Whatever repeats and stays harmless gets filed
as background — the refrigerator hum, the commute, eventually
the whole street.

This is useful. You would be overwhelmed if every hum, texture, face,
smell, and background sound felt as intense on the thousandth
encounter as it did on the first.

But the same machinery that filters refrigerator noises also
filters our lives.

- The person you love becomes familiar.
- Your home becomes familiar.
- Your body becomes familiar.
- The neighborhood becomes familiar.
- Even being alive becomes familiar.

And familiarity can masquerade as insignificance.

VISIBLE / PRESENT	
VISIBLE	tree, path, water
PRESENT	photosynthesis, groundwater, microbes, insects, wind, reflected wavelengths, cellular activity

NOTICE THIS

Choose one thing on a familiar walk.

Look at it as though you need to explain it to someone who has never seen it before. A tree. A storm drain. The edge of the lake. A bird. Your own hand.

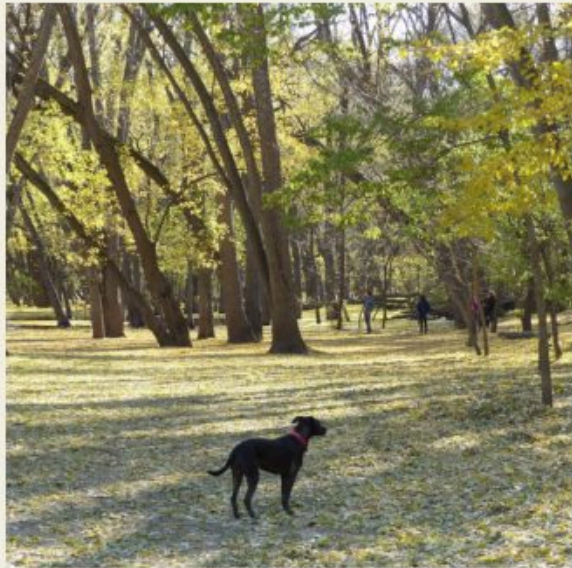


October 21, 2018, 2:58 p.m. / the leaves came down all week

THERE IS ALWAYS MORE



October 21, 2018, 3:11 p.m. / the route ran into water



October 21, 2018, 3:26 p.m. / same woods, twenty-eight minutes on

The extraordinary becomes ordinary not because it stopped being extraordinary, but because we stopped noticing it.

The Ordinary Is Not the Interruption

Walking is a good example.

You shift your weight forward. One leg catches you. Then the other. Your brain performs an enormous amount of calculation without asking permission.

You move across the ground while thinking about something completely unrelated.

It is an astonishing biological achievement. And almost nobody experiences walking as miraculous.

Unless they lose the ability to do it. Then suddenly crossing a room becomes precious.

This pattern appears everywhere.

Health becomes vivid during illness.
Home becomes meaningful when you're away.
A person's voice becomes precious after they're gone.

We often recognize value most clearly when something is threatened. Perhaps wisdom is learning to perform that restoration of attention before loss requires it. Not in a gloomy way. In a clarifying way.

Imagine seeing today's life through your own eyes thirty years from now. The face you see every morning. Your own body at its current age.

Would any of it still look ordinary? Probably not. It might look impossibly alive.

This doesn't mean forcing yourself to appreciate every second. That would become another form of self-improvement. Now you must optimize gratitude too. That misses the point.

Ordinary life includes irritation.
Boredom.
Laundry.
Work you don't want to do.
Arguments.

A meaningful life doesn't need every moment to feel meaningful. It may simply require understanding that these moments belong to the same life as the extraordinary ones.

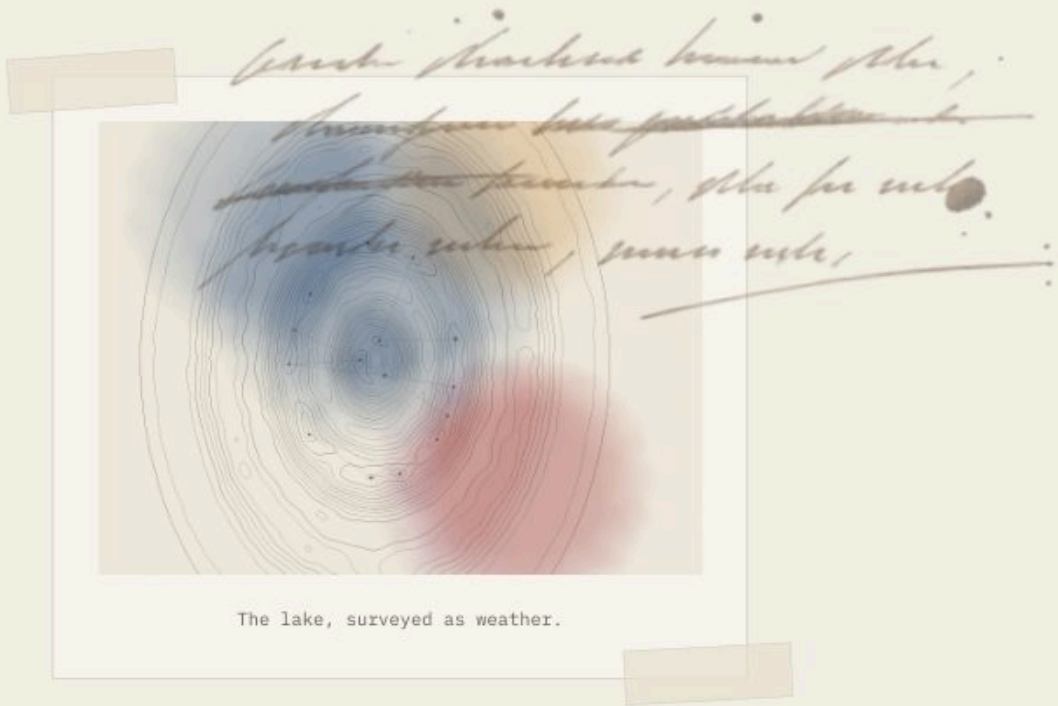
The boring afternoon isn't interrupting your life. It is your life.

Maybe this is why walking can feel different from other activities.

Walking doesn't necessarily accomplish much. You can walk in a circle and finish exactly where you started. Economically, almost nothing has happened.

But internally, things often rearrange themselves. Thought loosens. Attention widens. Weather, trees, people, smells, animals, and light start entering awareness again. The world resumes having texture.

The activity doesn't transport you out of ordinary life. It pushes you deeper into it.



Repetition gets a bad reputation. We imagine meaningful life as a sequence of new experiences.

But repetition may be what allows subtle differences to become visible.

On my usual route there is one tree I have photographed four times now — same tree, same angle, one photograph a season. Anyone watching would have seen a man standing in the street doing nothing. The tree never once repeated itself.

The light changes.
Birds arrive and leave.
A house gets painted.
Someone who used to walk beside you no longer does.

The route repeats. The world does not.

A routine is a stable frame around constant change.

The morning coffee is similar. You aren't. The neighborhood looks similar. It isn't. Even your body is continuously rebuilding itself.

The apparently static world is moving everywhere. Just slowly enough that distraction mistakes it for stillness.

A response that declines to something harmless repeating is called habituation, and it is one of the most widely observed forms of learning there is — described in animals with no brain at all. It also reverses. Change the stimulus and the response comes back.

We do the same thing with people. We turn them into categories.

Wife.
Husband.
Mother.
Father.
Friend.
Coworker.

The label makes someone familiar enough that we occasionally stop seeing the person underneath it.

But every person around us contains an entire private universe. Memories you'll never see. Things they notice when you're not there. Childhood versions of them-

selves. Internal conversations.

A person you've known for thirty years remains fundamentally mysterious. That mystery can disappear beneath routine.

Romantic love often begins with novelty. Everything about the person is interesting. How they laugh. What they eat.

Then years pass. The extraordinary person becomes the person standing in the kitchen. And maybe mature love is partly remembering that those are the same person.



The label, and the person underneath it.

There will always be extraordinary moments.
And they matter.

Travel somewhere beautiful.
Fall in love.
Make something you're proud of.
Have the adventure.
Celebrate the milestone.

Then you come down. And you make breakfast.

A good life isn't one containing the maximum number of extraordinary experiences.
Maybe it is one in which ordinary experience stops feeling like something you
need to escape.

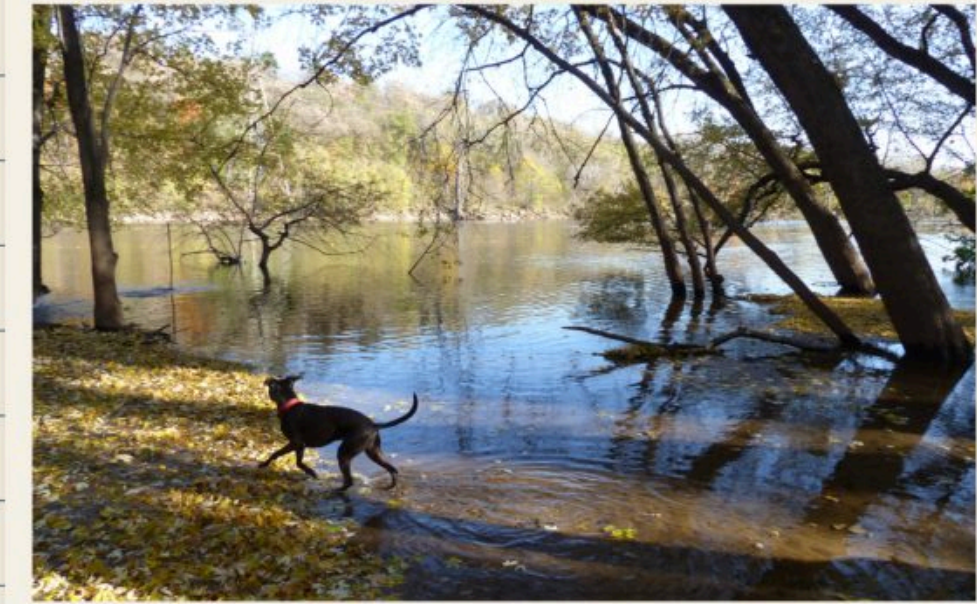
Because eventually, when we look backward, I suspect much of what we'll want
won't be another accomplishment.

One more dinner.
One more walk.
One more morning in the old house.
One more chance to hear somebody come through the door.
One more afternoon when nothing happened.

Maybe the ordinary days weren't the space between the important parts.

Maybe the ordinary days were the
important parts.

We just didn't know what they were worth yet.





The Secret Life of Attention

Attention is not a beam you point at the world. It is the thing quietly deciding what the world is going to be.

THERE IS A MAPLE ON THE CORNER OF MY STREET THAT I DID NOT SEE FOR ELEVEN YEARS.

I don't mean I never looked at it. I walked under it most mornings. I would have told you, if you had asked, that I knew the street well. Then one October afternoon I stopped in front of it, and it was enormous, and it had been enormous the entire time.

Nothing about the tree had changed. Something about me had.

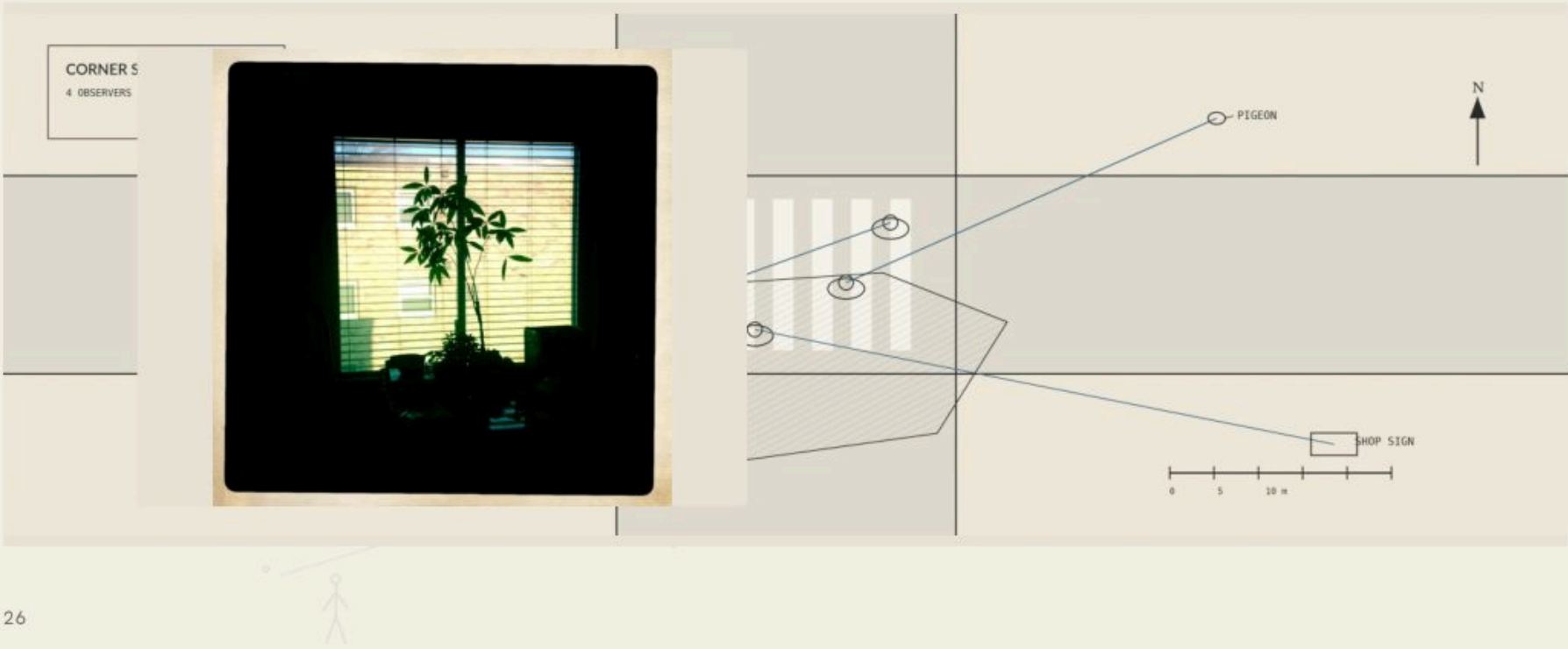
We talk about attention as though it were a flashlight. You point it, it illuminates, and the world sits there patiently waiting to be lit. The model is flattering. It is also mostly wrong.

Attention isn't a beam. It is closer to an editor. It decides, many times a second and almost entirely without consulting you, what gets to count as the world. The room you are sitting in contains more than you could ever process: the temperature of the air on the back of your hands, the particular pitch of the refrigerator, the fact that the light has moved four inches since you sat down. Your attention is not receiving that room. It is writing a very short summary of it and handing you the summary.

You have never once experienced a room. You have experienced a paraphrase.

Most of the time the paraphrase is excellent. It keeps you from walking into traffic and lets you find your keys. But it is also, quietly, the reason your life feels the way it feels, because the summary is the only version you are ever given.

Which raises a question I find genuinely uncomfortable. If attention is doing the editing, who is directing the editor?



For most of human history the answer was: your body, and the weather. Hunger. Cold. Movement at the edge of a field. Another person's face.

In the last twenty years an entire industry has been built on the discovery that the editor takes suggestions. Not orders. Suggestions. A notification is a suggestion. An autoplay is a suggestion. A feed is ten thousand suggestions arranged in the order most likely to be taken.

I don't think this is evil, exactly. I think it is the most successful engineering project of our lifetime, and I think it worked. The evidence isn't that we are distracted; distraction is ancient, and monks complained about it in writing. The evidence is that we have stopped noticing what we are distracted *from*.

There is a particular exhaustion that comes at the end of a day of total engagement with nothing. You put the phone down at eleven and cannot name one thing you actually attended to. It wasn't rest and it wasn't work. It was consumption without digestion.



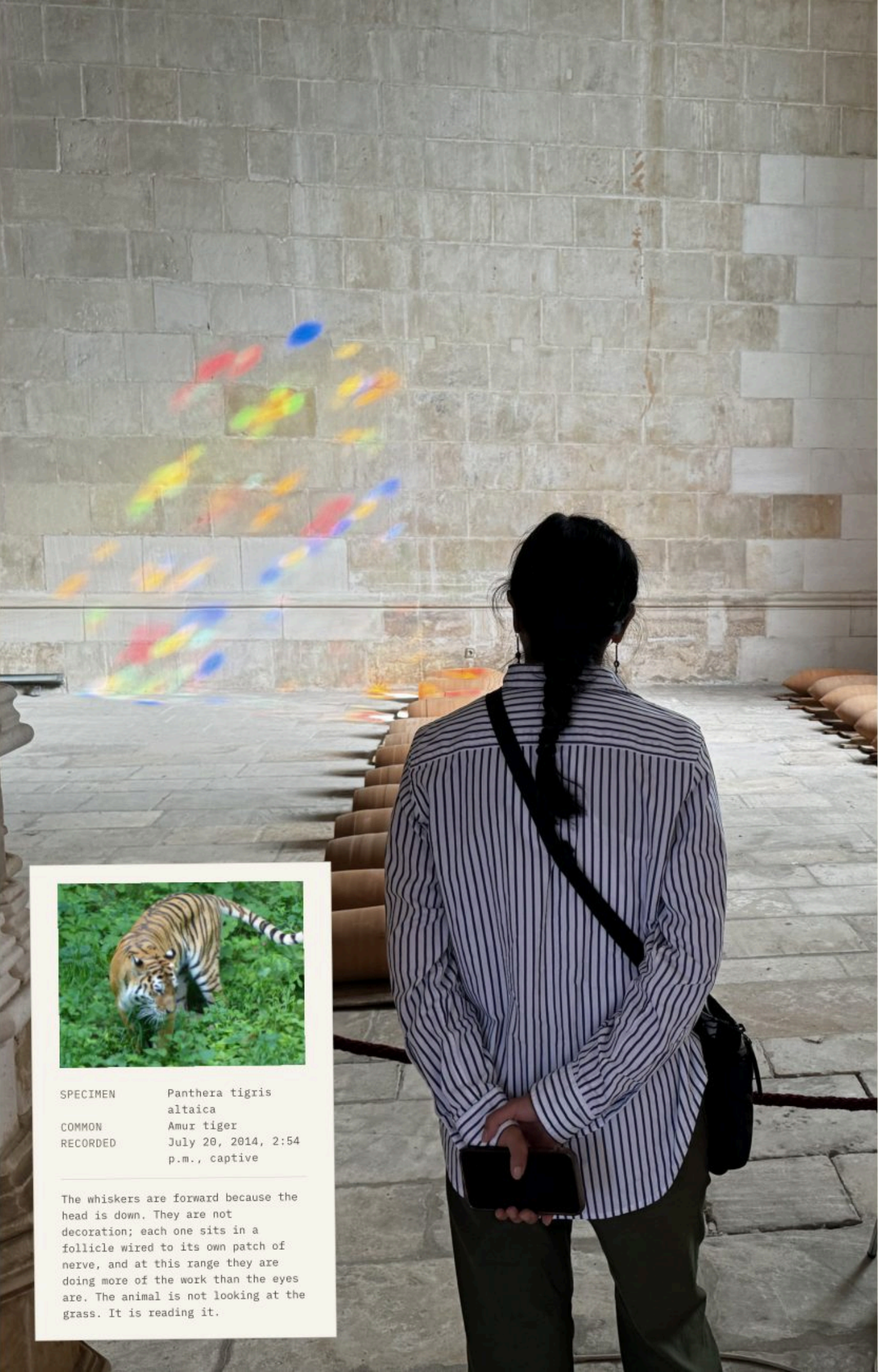
Large business-to-business enterprise employees (500+ employees)
Away from family
Away from hometown
Birthday in November
Console gamers
Engaged Shoppers
Facebook access (browser): Chrome
Facebook access (mobile): Android devices
Facebook access (mobile): Apple (iOS) devices
Facebook access (mobile): all mobile devices
Facebook access (mobile): smartphones and tablets
Facebook access (mobile): tablets
Facebook access (network type): WiFi
Family of those who live abroad
Frequent Travelers
Frequent international travelers

Friends of Men with a Birthday in 0-7 days
Friends of Men with a Birthday in 7-30 days
Friends of Recently Moved
Friends of Soccer fans
Friends of Women with a Birthday in 0-7 days
Friends of Women with a Birthday in 7-30 days
Friends of people with birthdays in a month
Friends of people with birthdays in a week
Friends of those who live abroad
Household income: top 25%-50% of ZIP codes (US)
Mobile network or device users
Potential mobile network or device change
Recent mobile network or device change
Uses a mobile device (25 months+)
Wi-Fi Usage

Meta Platforms, "Other categories used to reach you". Personal data export generated 19 August 2026. Reproduced in full, unedited.

You have never
once experienced
a room. You have
experienced
a paraphrase.

Plate 3. A bare wall, a window out of frame, and a scatter of colored light between them. She has the room to herself, and this is what she is looking at.



SPECIMEN	Panthera tigris altaica
COMMON	Amur tiger
RECORDED	July 20, 2014, 2:54 p.m., captive

The whiskers are forward because the head is down. They are not decoration; each one sits in a follicle wired to its own patch of nerve, and at this range they are doing more of the work than the eyes are. The animal is not looking at the grass. It is reading it.

Here is the part I keep returning to. Attention is trainable, but not by force.

You cannot decide to notice more the way you decide to answer an email. I have tried, with the enthusiasm of a man who solves things. Concentration can be forced; it responds to deadlines and coffee and fear. Noticing cannot. It arrives or it doesn't. What you can do is arrange the conditions in which it becomes likely, and nearly all of those conditions turn out to be versions of the same thing, which is slowness.

Walking works. Sitting still works. Ten minutes in front of one painting works better than a museum. Boredom works, though nobody wants to hear it. Grief works, unfortunately, and thoroughly. So does a dog, who will stop at the same square yard of grass every day for nine years and never once find it uninteresting.

What all of these have in common is that they lower the speed of the world until the detail can catch up with you.

The maple was not a revelation. It was the ordinary result of standing still for ninety seconds on a street I usually crossed in twenty.

ten minutes, one wall



FIELD NOTES

The rope protects the amphorae. What she is watching needs no protection.

The window throwing the color is somewhere behind her. She has not turned to find it.

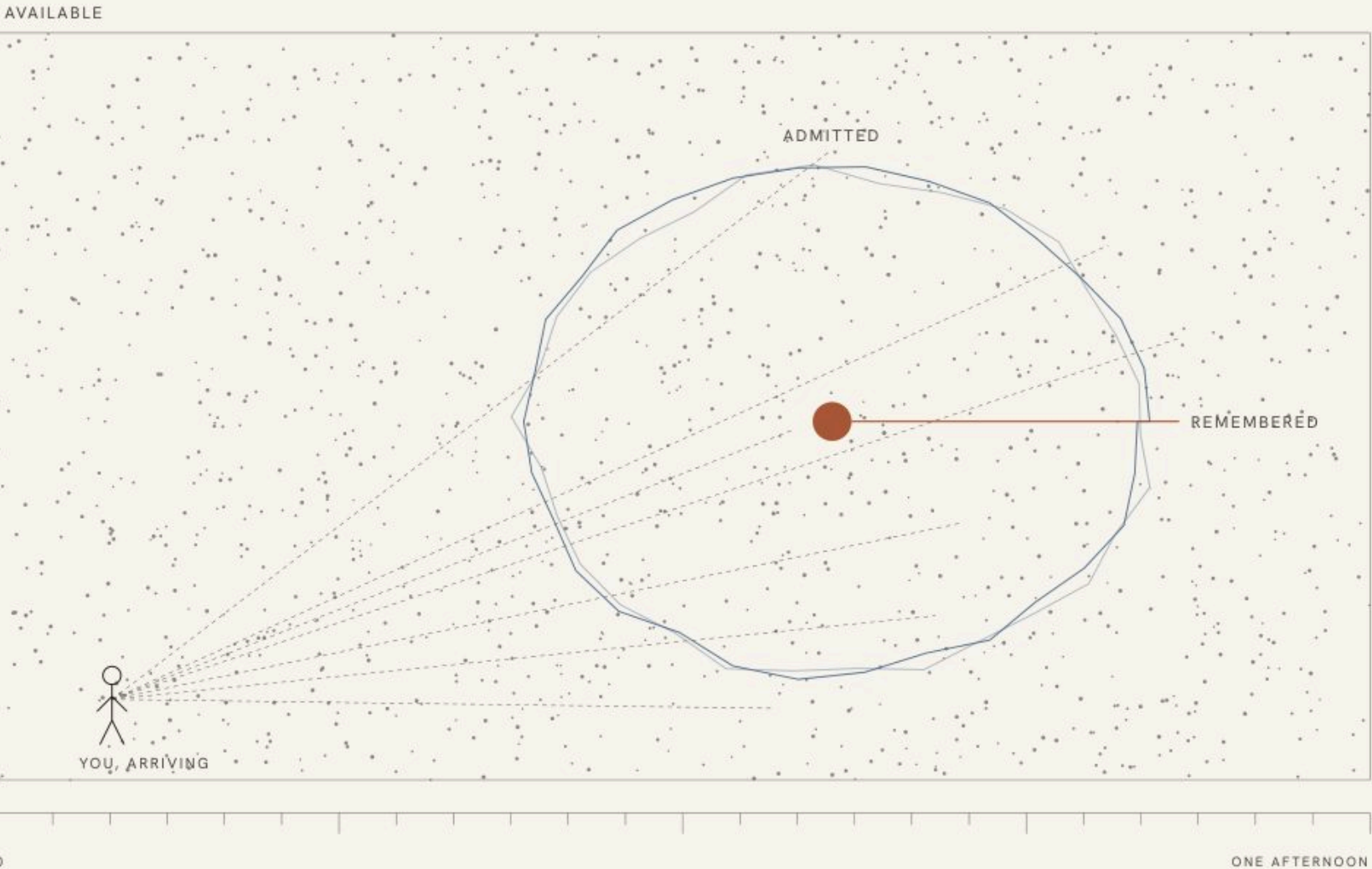
The phone is in her hands, behind her back.

Standing still is the whole technique. In an hour the wall will be bare again.

FIGURE 02.1

What the editor kept

Almost everything that happens near you is discarded before you meet it. This is not a flaw. A version of you that admitted the whole room would be unable to cross a street. But the ratios are worth looking at once, because something like them is the shape of a life.



Available. Everything the room was actually doing while you were in it.

Admitted. The fraction that reached awareness at all.

Remembered. The fraction you will still have tomorrow.



William James, writing more than a century before any of this was engineered, argued that a life is made of whatever a person consents to notice. He meant it as psychology. It reads now as an inventory.

Attention is the only genuinely nonrenewable thing you have. Money comes back. Energy comes back. Even time comes back, in the sense that tomorrow arrives whether or not you have earned it. Attention does not work that way. Every minute of it is spent in the moment it is spent, and the sum of those minutes is not a metaphor for your life. It is your life, in the most literal accounting available.

This should probably produce anxiety. Mostly it produces the opposite in me.

If attention is the currency, then the price of a good day is far lower than I was led to believe. I have had genuinely extraordinary experiences that I barely registered. Arranged at expense, photographed carefully, and gone. I have also had ten minutes on a back step with a cup of coffee, watching a bird do something idiotic in a gutter, that I can still see with total clarity a decade later.

The difference was not the quality of the events. It was whether I showed up for them.

We spend enormous effort trying to make our lives more interesting. Very little of it goes into becoming more interested. These are not the same project, and only one of them is available this afternoon.

NOTICE THIS

Walk five minutes of a route you know by heart. Find one thing you have never seen before. It is there. It has been there the whole time.

The world didn't change. Your attention did.

I think about that most days now, usually badly, usually while checking something I do not need to check. But I know where the maple is, and some mornings I look up.

Eleven years is a long time to walk under something beautiful.

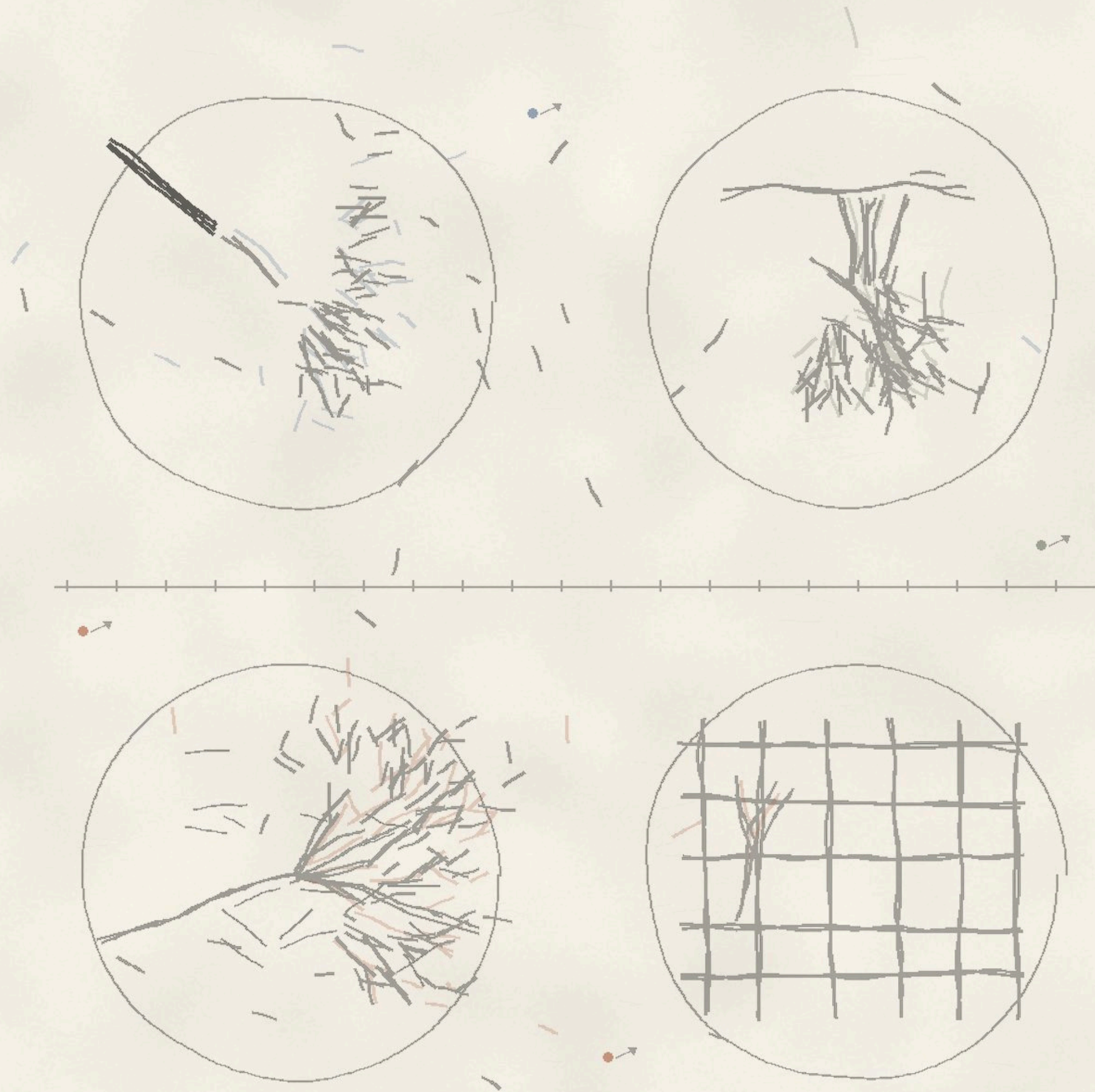
It is not, as far as I can tell, too long.





nothing happening

I had nearly walked past before the small movements separated themselves from the view. The mooring lines lifted, slackened, and pulled tight again. Water pressed against the dock in short runs. A loose line tapped one mast, stopped, then started after the next wave. The boats shifted by inches and returned. I stood there long enough for the same motions to repeat without becoming identical. The bright patch beneath the hulls kept breaking into pieces and joining itself again.



— NOTICE THE PATTERNS

Part II

What Are We?

The order was never installed. It is
being held in place, continuously,
by things that cannot see it.

Flocks, networks, nests, and the intelligence that is not in your
head.

- 01 The Beauty of Systems Nobody Designed
- 02 The Intelligence Outside Your Head



The Beauty of Systems Nobody Designed

Order can arrive without a plan when many small parts keep answering one another.

A COLONY WORKS BEHIND GLASS AT THE END OF THE BARN. EACH BEE STOPS AT CELL AFTER CELL, TOUCHES THE RIM, AND MOVES ON. ONE BACKS OUT AND REVERSES DOWN THE COMB. TWO MEET HEAD TO HEAD, PASS ANTENNAE, AND SEPARATE. FROM STANDING HEIGHT THE SHEET LOOKS PLANNED. CAPPED HONEY IS BANKED ALONG THE TOP, THE BROOD IS HELD IN THE WARM MIDDLE, AND THE EMPTY CELLS WAIT AT THE EDGE. THERE IS NO BEE AT THE FRONT HOLDING A DRAWING. NONE CAN SEE THE WHOLE COMB. THE WALL EXISTS BECAUSE THOUSANDS OF SMALL BODIES KEEP WORKING THE SAME SURFACE AND ANSWERING WHAT THE LAST ONE LEFT. THE WORKED COMB DARKENS. THE UNUSED EDGE STAYS PALE. WINTER WILL DRAW THE STORES DOWN. A KNOCK WILL CRACK IT. THE COLONY WILL BEGIN AGAIN FROM WHAT REMAINS. BY AFTERNOON, A SYSTEM IS RUNNING WHERE THERE WAS ONLY WAX, HUNGER, AND REPEATED CONTACT.

We are trained to look for the maker. A nest suggests an architect. A flock suggests a leader. A network suggests a plan drawn somewhere before the first connection was made. This habit works well for tables and bridges. It fails in much of the living world. The order appears first in the interactions. Each part follows a narrow set of conditions. Nearness is maintained. Collisions are avoided. A turn

spreads when nearby bodies turn. Mud is placed where nearby structure and scent make placement more likely. No part understands the final form. There is no final inspection and no moment when the system sees itself completed. The form is what those instructions become when they are repeated by many bodies over time.



A flock of European starlings, *Sturnus vulgaris*, can turn through the winter sky as if it were one dark sheet being folded. The birds do not all watch a single leader. Each responds to a small number of nearby birds. A turn begins somewhere and travels across the flock faster than any one bird could inspect the whole shape. The pattern has no fixed center. Birds enter, leave, trade positions, and continue to produce a coherent edge.

The flock can stretch around a tree, close an opening, and divide into two moving bodies without suspending the rules that hold it together. What looks like an object is a temporary agreement among moving animals. It lasts only while the local corrections continue. The flock is not controlled from above. It is held together from beside.

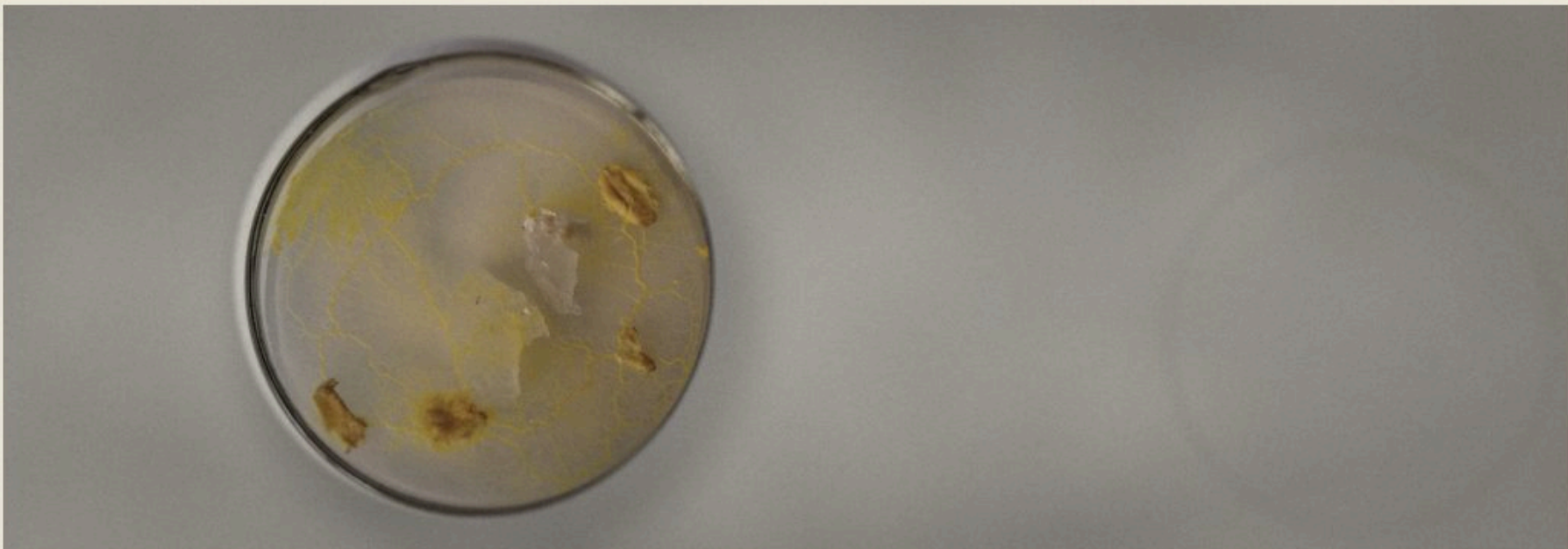


This one did have an architect.

In a 2008 field study, European starlings in a large flock were found to interact with about six or seven nearby birds, whatever the physical distance between them. A turn can cross the whole flock through local updates, with nobody giving an order.

The same logic can be seen in a yellow slime mold spread across a laboratory dish. *Physarum polycephalum* has no brain and no nervous system. It extends thin tubes toward food, reinforces the routes that carry material efficiently, and withdraws from routes that do not. Researchers placed food at positions corresponding to cities around Tokyo. The organism built a network with costs and resilience

comparable to the rail system. Thick tubes remained along useful connections. Redundant branches thinned and disappeared. The resulting network kept more than one route to important points without maintaining every possible route. It did not know Tokyo. It did not know railways. It answered gradients of food, light, and its own internal flow. A useful map emerged from a body solving only the conditions touching it.



The subterranean nests of the African termite *Apicotermes lamani* form through another set of local acts. A worker carries a pellet of mud and places it in response to nearby structure and chemical cues. Other workers add nearby. The new surface changes where bodies can move and where later material is likely to be placed. Floors extend, and ramps connect them to

chambers. Architecture and behavior keep revising one another through the work itself. No termite measures the nest from one side or compares it with a finished model. A labyrinth used by the whole colony emerges from interactions among termites, pheromones, and mud. It is not the execution of a hidden blueprint. It is a conversation conducted in material.



In a 2010 experiment, *Physarum polycephalum* was given food at positions matching cities around Tokyo. The network it built balanced cost, transport efficiency and fault tolerance at levels comparable to the rail system.

Figure 2. A muqarnas vault at the Alhambra: thousands of small cells stepping inward to one form. The essay's exception — this ceiling had a diagram.



SPECIMEN Danaus gilippus
COMMON Queen
RECORDED July 20, 2014, 4:46
 p.m., captive

Milkweed as a larva, cardenolides carried into adulthood, and the same warning pattern the monarch wears. Neither species is copying the other. A predator that learns one avoids both, so the pattern is held in place by the birds. How distasteful a queen actually is depends on the plant it grew up on.

The honeycomb of *Apis mellifera* looks drafted. Its repeating cells divide a surface into chambers of similar size. The geometry tempts us to imagine a diagram passed through the hive. The bees have something smaller. New cells begin in the grooves between cells already built. Each one inherits its position from the neighboring work. The cells begin closer to cylinders, then become hexagonal through construction behavior and physical shaping that researchers still debate. The regularity is real, but it is not perfect. Cells bend near the edge. Rows adjust around obstacles. A misplaced cell changes the shapes around it. The comb is exact where exactness helps and flexible where the material requires it. Its beauty does not come from escaping those limits. The limits are what give the pattern its shape.

FIELD NOTES

Ants cross the seam between two paving slabs and tighten into a single line.

A starling flock compresses above bare trees, then opens at one edge.

Rainwater splits around a leaf and rejoins below it.

The honeycomb changes angle where it meets the wooden frame.

this one had a foreman





The path was poured for other feet.

These systems are often called self-organizing, though the phrase can make them sound independent of the world around them. They are not. Water follows slope. Roots follow moisture and gaps in soil. A river basin takes its branching shape from rock, rain, gravity, vegetation, and the channels cut by earlier water. The map of a watershed is a record of resistance as much as flow. A harder ridge remains because water moved elsewhere. A new channel deepens because earlier water made the next passage easier. The pattern is not imposed on the materials. It records what the materials have encountered. When soil changes, roots change. When one channel closes, pressure moves elsewhere. Order is not a finished arrangement. It is the visible history of many constraints.

A human body is full of the same kind of order. Cells exchange signals and repair damage without waiting for a conscious instruction. The bacteria in the gut digest compounds human enzymes cannot break down. Blood vessels widen near working muscle, while an immune response recruits cells through chemical signals released at damaged tissue. The response becomes coordinated before the conscious mind knows the site is injured. None of this makes the body simple or harmonious. Local rules can produce inflammation, tumors, scars, and habits that outlast their usefulness. Emergence is not another word for goodness. It names the way large consequences can arise from small processes that have no view of the whole.

A 2016 estimate for a seventy-kilogram reference man counted about thirty trillion human cells and thirty-eight trillion bacterial cells. The ratio varies between people, but the two populations are the same order of size.

NOTICE THIS

Stand beside one moving system for five minutes. Watch only the nearest interactions. Mark the moment a pattern appears without anyone announcing it.

Human systems do this too. A footpath appears where enough people cut the same corner. A language shifts because millions of speakers shorten sounds, borrow terms, mishear phrases, and repeat what survives. A city grows around old property lines and routes to water. Later transit lines inherit those earlier choices. Traffic gathers on a street because drivers avoid another street, then moves again because enough drivers make the same correction. Nobody designed the total result, even when thousands of people designed pieces of it. The system carries intention without having a single intention of its own. That is why it can be useful, wasteful, durable, and difficult to change at the same time. The newest of these footpaths is worn by glances instead of feet, and nobody designed its total result either.

A line of ants crosses the pavement, thinner where the surface has cooled. A bicycle passes through it. For a few seconds the route is broken. Ants circle the gap and touch the ground with their antennae. Some return to the curb. Others find the trace on the far side. The line reforms, not exactly where it was, but close enough to reach the same opening. Nothing announces that the repair is complete. There is only movement where movement had stopped. By morning, a different obstacle may bend the route somewhere else. The pattern will persist by changing. The system survives because no single part contains it. It is distributed among the bodies, the pavement, the scent, and the next response.

**The next pattern is already taking shape
at the edge of what each part can sense.**

**Order is sometimes
a verb performed by
many small things.**



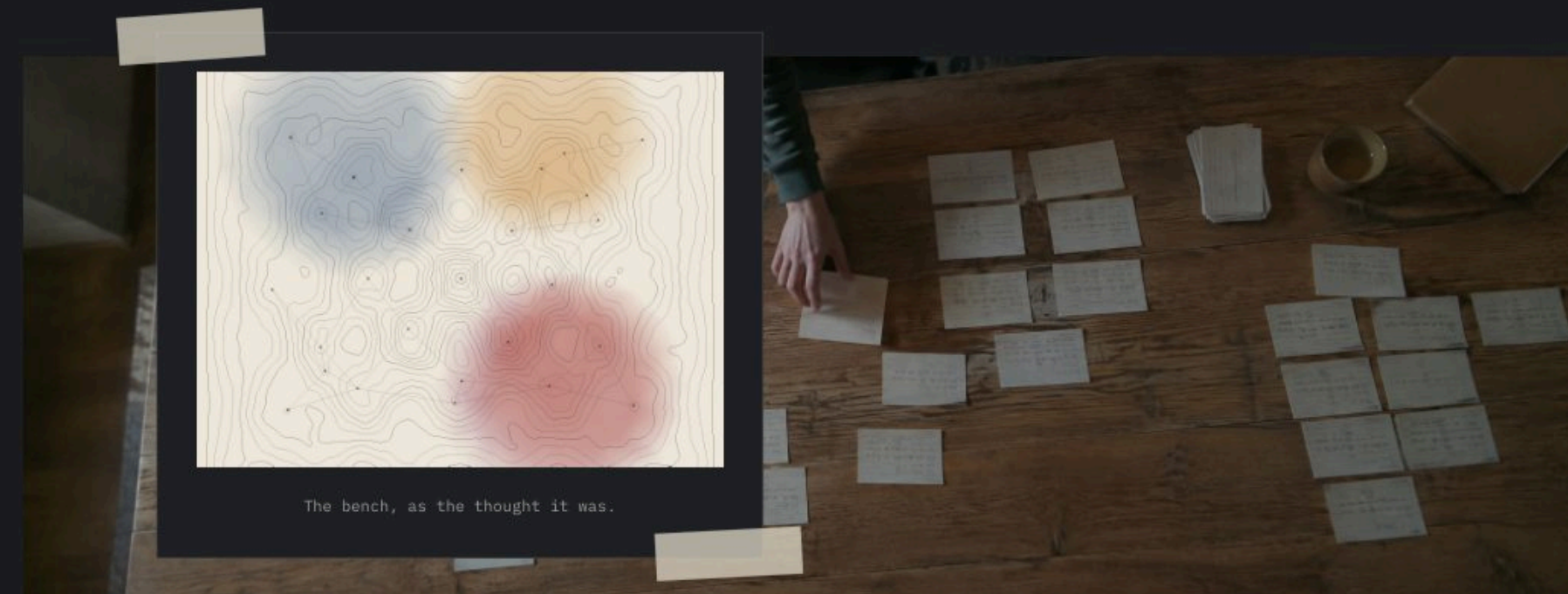
The Intelligence Outside Your Head

Thinking becomes larger when the world is allowed to carry part of the load.

A PENCIL LINE CROSSES A PINE BOARD AT THIRTY-SEVEN AND A QUARTER INCHES. THE TAPE MEASURE SNAPS SHUT. THE NUMBER LEAVES WORKING MEMORY ALMOST AT ONCE, BUT THE LINE REMAINS. A SAW CAN NOW BE PLACED AGAINST IT WITHOUT THE MEASUREMENT BEING REPEATED. THE BOARD IS HOLDING PART OF THE TASK. THE MARK IS NOT INTELLIGENT BY ITSELF. IT DOES NOT KNOW WHAT IS BEING BUILT. STILL, THE CUT WOULD BE LESS RELIABLE WITHOUT IT. THE THOUGHT HAS BEEN DIVIDED BETWEEN A NERVOUS SYSTEM AND A SURFACE. ONE REMEMBERS THE INTENTION. THE OTHER PRESERVES THE LOCATION. TOGETHER THEY CAN DO SOMETHING NEITHER COULD MANAGE AS WELL ALONE. THE ORDINARY WORKSHOP IS FULL OF ARRANGEMENTS LIKE THIS. A STOP BLOCK REPEATS A LENGTH. A JIG PRESERVES AN ANGLE WHILE A ROW OF PARTS SHOWS WHAT HAS BEEN FINISHED.

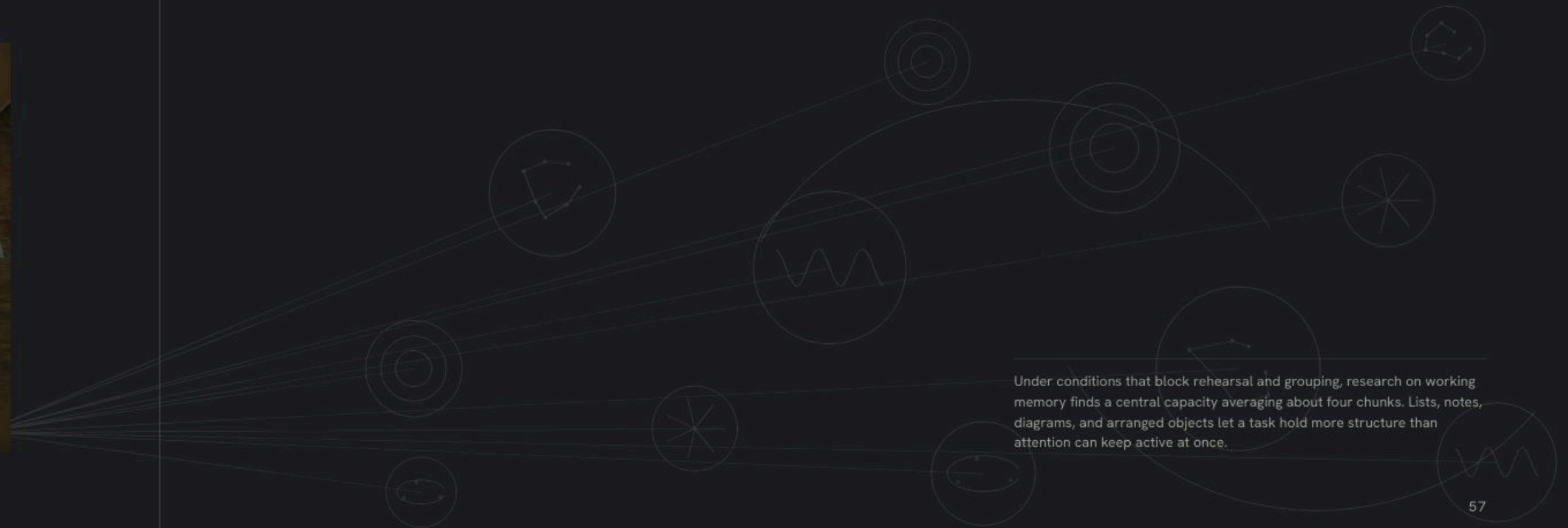
Human attention is narrow. Under controlled conditions, the active part of working memory often holds about four chunks at once. The exact number changes with the task and with what counts as a chunk, but the limitation is plain in daily work. A long calculation becomes unstable when every intermediate result must be remembered. Put the figures on paper and the problem changes. Earlier steps

remain available while attention moves forward. A diagram can preserve relationships that would dissolve if they had to be rehearsed as sentences. Scrabble tiles can be moved until a word appears. Index cards can be grouped before the categories have names. The hands make a possibility visible, then the eyes return it to the brain. Thinking continues through the loop.



Tools do more than increase force. They carry instructions. The bubble in a spirit level turns gravity into a readable position. A kitchen scale converts pressure into a number that can be compared with a recipe. The shape of a key preserves a solution to a lock. None of these devices decides the larger purpose. Each narrows a difficult judgment into a condition that can be

checked. Previous minds are present in the shape of the tool. Someone learned where a handle should bend, which edge needs a guard, and how much tolerance the material allows. That learning becomes available to a later user who may never know the history. The tool is a small archive made operational.



Under conditions that block rehearsal and grouping, research on working memory finds a central capacity averaging about four chunks. Lists, notes, diagrams, and arranged objects let a task hold more structure than attention can keep active at once.

Navigation shows how easily intelligence changes location. A person who knows a city carries intersections, landmarks, and likely routes in memory. London taxi drivers spend years building this internal map, and brain imaging has found structural differences associated with that work. A driver following satellite directions performs a different task. The map is outside. Position is calculated elsewhere. The driver must keep

the car in its lane, read the next instruction, and match the screen to the street. Less route knowledge is required, but the journey still depends on a distributed system of satellites, clocks, road data, and signals received through glass. The intelligence has not vanished. It has moved into an arrangement that no traveler owns.



Groups have always stored knowledge this way. One person remembers the names. Another remembers the sequence. A surgical team does not rely on one mind to contain the whole operation. Instruments are counted aloud. Readings remain visible. Roles are assigned so that one person's attention can catch what another misses. A ship's position once emerged from charts, timed observations, and several people checking the

same figures. Modern control rooms use different screens, but the principle remains. Reliable work is often produced by a system that distributes memory and verification across people and objects. Expertise still matters. The expert is partly the person who knows where the knowledge lives, when to call for it, and how to recognize a result that does not fit.

In a 2000 MRI study, licensed London taxi drivers had more gray-matter volume in the posterior hippocampus than controls. The difference correlated with years spent navigating, evidence that repeated use can alter the brain's spatial system.

Figure 3. A bench left ready. None of the thinking is in the tools, and none of it happens without them.



SPECIMEN Varanus komodoensis
COMMON Komodo dragon
RECORDED July 20, 2014, 4:12
p.m., captive

The tongue is forked so that it can sample two places at once. Both tips press to paired organs in the roof of the mouth, and the difference between the two readings is the direction. The animal is not picturing the field. It is standing in it and taking the difference.

Computers make this distribution harder to notice because retrieval can feel immediate. A name, date, or method appears after a few typed words. Experiments have found that when people expect information to remain available, they may remember where it can be found better than the information itself. This is often described as forgetting, but it is also a change in strategy. Couples do it. Work teams do it. One person becomes the place where birthdays are kept. Another knows the passwords or the route through a bureaucracy. The shared system can hold more than any member, though it creates new weaknesses. When access fails, the missing knowledge becomes visible. The blank search field reveals how much had been stored beyond recall.

FIELD NOTES

A pencil line crosses a board a few millimeters beyond the saw blade.

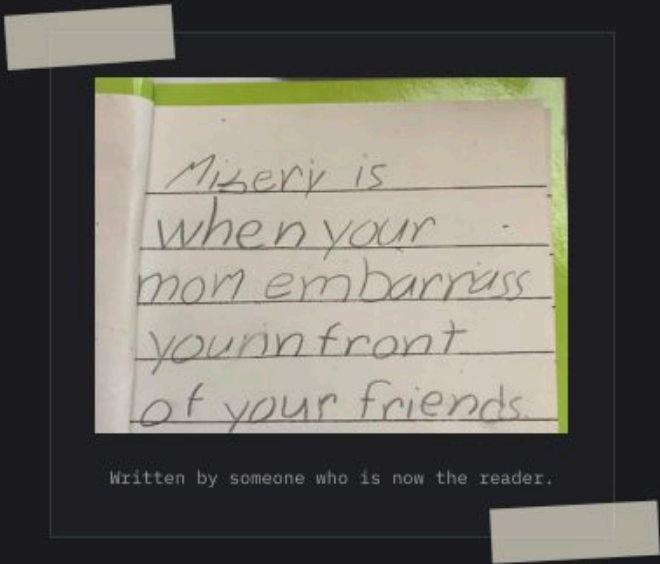
Four loose photographs change order on a table until two belong beside each other.

A driver glances from the road to a blue route line and turns before the street name appears.

The spider rests at the center of its web with one foreleg on a radial thread.

held by the room





Language is the oldest infrastructure of this kind. A word compresses distinctions learned by people who lived before the current speaker. Technical vocabulary can carry a procedure across decades. A place-name keeps an earlier use attached to a landscape after the building is gone. Writing makes the effect durable. Instructions survive their author. A margin note can argue with a reader born later. Libraries do not merely contain thoughts that once happened inside individual heads. They allow new thoughts to be assembled from records that no single person could have produced or retained. The intelligence belongs to the living reader, but the materials arrive from outside. Thinking is inheritance put back into motion.

The arrangement can extend beyond human artifacts. An orb-weaving spider sits with its legs touching radial threads. Vibrations from prey, wind, or another spider travel through the web. The spider can alter thread tension and change how those signals move. The web is not only a trap. It is an adjustable sensory surface built outside the animal's body. Part of the work of detection happens in silk before a signal reaches the nervous system. The boundary between organism and instrument becomes difficult to draw cleanly. The spider made the structure, then the structure changes what the spider can perceive. Body and environment keep revising the same act.

Four experiments published in 2011 found that people expecting later computer access recalled the location of information better than the information itself. Memory had not disappeared. Part of the task had shifted toward retrieval.

NOTICE THIS

Put one unfinished problem on paper. Move its pieces until the next step becomes visible. Notice which part of the answer appears in the arrangement rather than in memory.

External intelligence is powerful because it is vulnerable. A notebook can be lost, and a navigation database can be wrong. A checklist copied without understanding can preserve an old mistake with impressive consistency. Systems that make action easier can also hide the knowledge needed to question them. The answer is not to keep everything in memory. That has never been possible. The useful distinction is between support and surrender. A good external system leaves enough of its workings visible that a person can detect failure. The pencil line can be measured again. The map can be compared with the road.

The cut board is carried to the frame and does not fit. The line was clear, but the measurement was wrong. The builder returns to the tape, the drawing, and the opening where the board belongs. Each holds a different part of the error. The corrected length emerges from comparing them. No single object contains the answer. No single glance completes the work. Intelligence appears in the traffic between memory and material. The mind reaches outward, receives resistance, and changes course. The room is not a backdrop for thought. It is one of the places where thought happens. Every tool so far has held the thought it was given. The next one answers back.

Restore them and the unfinished work begins again.





never been lost

The tag on her collar has a name on one side and a square code on the other. She was asleep when I checked it. The code opens a page with her photograph and a form for whoever finds her, and a note that says she is friendly, which is true. She has never been lost. The tag clicks against her bowl when she drinks. I scanned it once, standing in the kitchen, to see what would happen. Somewhere a server confirmed that she was not missing.



A dog with a URL.



— EXPAND THE APERTURE

Part III

The World Is Changing

You are inside the change,
not after it. That is the only
vantage point that ever exists.

Machines that are learning to think, and a world that has already
stopped being the previous one.

- 01 The Strange Privilege
- 02 The Last People Who Remember Waiting



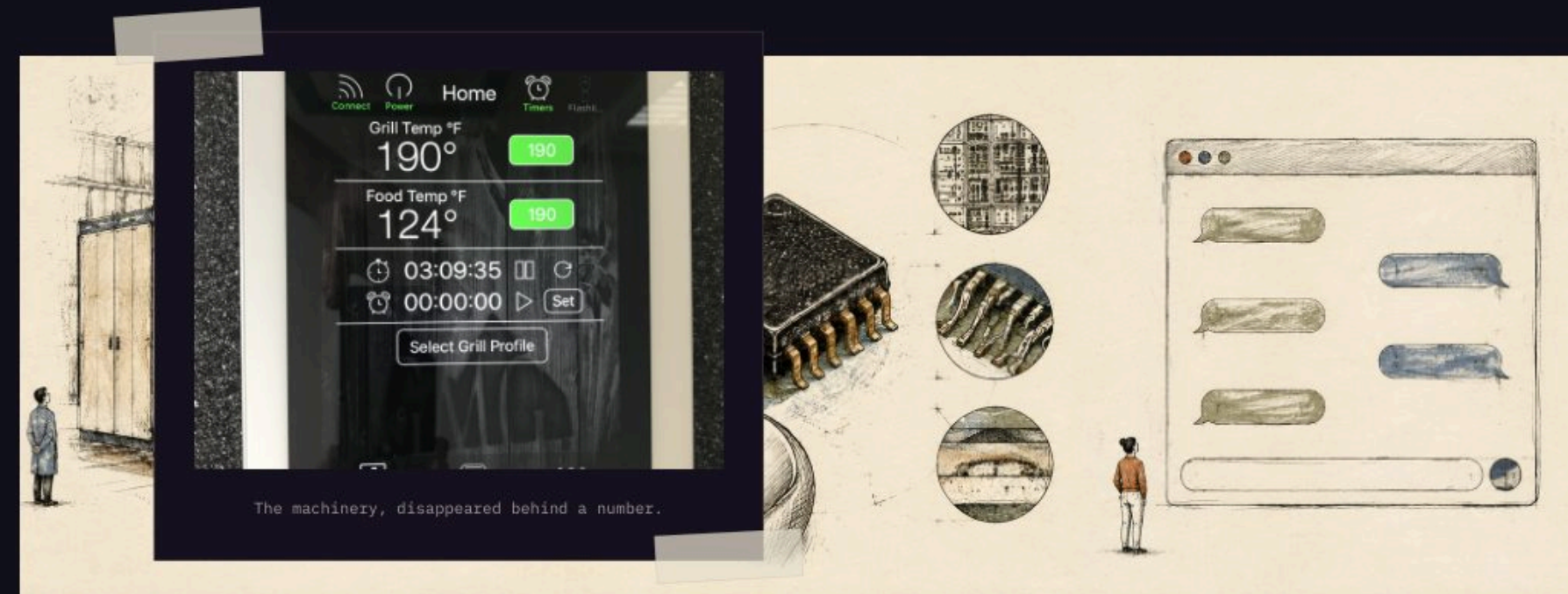
The Strange Privilege

The rare condition is not living after a transformation, but living while its rules are still unsettled.

A BLANK FIELD SITS IN THE MIDDLE OF A SCREEN. A SENTENCE IS TYPED INTO IT. THE MACHINE PAUSES, THEN BEGINS RETURNING WORDS. THEY ARRIVE AT THE SPEED OF THOUGHT BUT WITHOUT THE HESITATIONS OF A PERSON DECIDING WHAT TO SAY. SOME ARE USEFUL. ONE CLAIM IS WRONG. THE TONE IS ALMOST RIGHT. A PARAGRAPH THAT MIGHT HAVE TAKEN AN HOUR NOW EXISTS IN LESS THAN A MINUTE AND STILL NEEDS TO BE READ WITH CARE. NOTHING IN THE ROOM LOOKS HISTORIC. THERE IS A MUG BESIDE THE KEYBOARD AND DUST ALONG THE MONITOR STAND. THE EVENT RESEMBLES OFFICE WORK. THAT IS HOW A TECHNOLOGICAL REVOLUTION OFTEN ENTERS DAILY LIFE. IT DOES NOT ANNOUNCE THE FUTURE. IT OFFERS TO FINISH A TASK BEFORE LUNCH.

Later generations inherit revolutions as furniture. Electric light becomes a switch on the wall. Recorded music becomes a button. Satellite navigation becomes a blue line that appears before the car moves. The machinery disappears behind a reliable action, and the long period of uncertainty is compressed into a date in a textbook. Living through

the change feels different. Standards conflict. Devices fail in public. Old skills remain necessary while new ones become unavoidable. People argue about names because the categories have not settled. A tool can be called a toy for years, then become the place where ordinary work begins. The transformation is visible only in fragments, one convenience at a time.



The fragments accumulate. In 1971, a chip smaller than a fingernail started computation on its long walk out of the specialized rooms — into calculators, then appliances, then cars. No single household object contained

the revolution. A machine stopped being one machine. It became a component that could disappear inside another object and quietly change what that object was able to do.

Intel introduced the 4004 microprocessor in 1971 with 2,300 transistors on a chip smaller than a fingernail. Computation that once occupied a room had begun its long movement into ordinary objects.

ADVERTISERS HOLDING YOUR ACTIVITY OR INFORMATION7,325 unique · 154 shown

Uniagency	Weber Grills	Wildsam Field Guides	Doctoralia México	MercadoLibre.com -	GCommerce
LEGO	Disney Streaming	Serpentine Account	株式会社電通デジタル (Dentsu	Celulares y Teléfonos	BM350 Whiter XMDN USA 8
Harper's Bazaar	MAXIMILIAN EQUESTRIAN	Realtor.com	Digital Inc.)	Lenovo X Dentsu	Foursquare City Guide
Instacart	TBG	MediaCom Zagreb	MioDottore	Jameda Pro	Briggs & Caldwell
ELLE Taiwan	Sonos	Weber Barbecues	Better World Books Account	Performics	Brisket Chips Co.
JustWatch	Huntington National Bank	Australia/New Zealand	iProspect	MPlatform Romania	Sojern
Adobe	Virgin Voyages	Domestika	AKA - Test	RingCentral	My Health Teams
Vuori	Perplexity AI	Niche	Adbid - Marketing Digital	BuzzFeed Branded	Synchrony
Kiara Sky Professional	Dyson	Peter Millar	Paycom	Distribution	Vail Resorts
Nails	King	Ubisoft Global Publishing	Accenture M+C Media - UK	That Lot	Best Buy
Non ddtc 1 zocket manager	Audible	Doctoralia Brasil	Advertising Srbija	Sabre Corporation	Starcom USA
IKEA	AccessiBe	Orthofeet	Lenovo Italia	IG Ads Identity	Willand Data Tactics
Silhouette Eyewear	GroupM Media India	UM MELI - CO	Zorba's Coney Island	Direct Relief	Chewy
Elle.it	Initiative Media South	Topo Designs	Strike Social	GroupM Danmark	iProspect Detroit
Paramount Pictures	Africa	DoktorTakovimi.com	Expedia	Sun Country Social	Pixel Motion
Disney	Vrbo	IKEA Canada	Digital Advverting 2024	Neko Test Page	DXL Big + Tall
A24	Hearts & Science	Doctoralia España	MC SG	BJ's Wholesale Club	Hayabusa Fight
Lionsgate	Kohl's	IKEA 이케아	Vamp	OMD USA	Acxiom
EM NW EURO	ZnanyLekarz	Doctoralia Colombia	Meetsocial HK Digital	WMX	Green & Wood Media
Reality Labs	Meta	Assembly APAC	Marketing Co., ltd-1	Epsilon Audience Data	Services
Yuguo用果跨境电商	GVO 1 Zocket Manager	Sioux digital 1:1	Adobe Photoshop Elements &	Provider	LaunchBoom
Crate and Barrel	ELLE España	Life360	Premiere Elements	OpenTable	SupplyX
Hearst Netherlands	MediaCom USA	Lenovo EMEA Marketing	Oracle Data Cloud	Brand survey	First Avenue & 7th St
Intuit TurboTax	HMS Global - GBP	Publicis Media Indonesia	Drive Toyota	Drive Toyota	Entry
LiveRamp	Suno	GipoNext	RPA Advertising	RPA Advertising	Ticketmaster
Cosmopolitan UK	IKEA Austria	Chris May 1	Lumenad	Lumenad	
ELLE Japan	iHeartRadio	YETI	Take-Two Interactive	Take-Two Interactive	
KHM	Chris May 2	Peacock TV	Hulu	Hulu	

Meta Platforms, "Advertisers using your activity or information". Personal data export generated 19 August 2026. Of the first 160 entries in export order, 154 are printed here unedited and unsorted; six are withheld. The 7,165 beyond them are not printed because they would fill this page forty-six times over.

A revolution is
easiest to miss
when it arrives
as a convenience.

Learning systems make a different movement. Traditional software follows rules written in advance. A learned model is shaped by examples. During training, it adjusts internal values until its outputs fit patterns in the data closely enough to be useful. The resulting behavior can be difficult to reduce to a short list of instructions. In 2017, a research paper about machine

translation described a better way for a system to weigh words against one another. The method proved useful far beyond the problem it was written for. Language became one of the materials that large computational systems could model, continue, classify, and rearrange.



The public encounter with this change has been unusually direct. Earlier advances in machine learning often arrived inside ranking systems or fraud detection. Their decisions affected daily life without presenting themselves as collaborators. The editor that decides what gets noticed had, by then, acquired account managers. Generative systems appeared in a blank field. A person could ask for a paragraph or a piece of

code and watch an answer form. The interface made a research trajectory feel conversational. It also made the weaknesses easy to meet. The same system that produces a clear explanation can invent a source. An image can preserve a difficult lighting condition and misunderstand a hand. Fluency and reliability arrive as separate properties. The user has to learn the difference.



The 2017 transformer paper described a network based on attention rather than recurrence. Its larger model trained for 3.5 days on eight P100 GPUs and set new reported results on two translation tasks.

Figure 4. The dog under the desk. Whatever is happening above her is, from the floor, a Tuesday.



SPECIMEN	Anodorhynchus hyacinthinus
COMMON	Hyacinth macaw
RECORDED	July 20, 2014, 1:29 p.m., captive

The beak opens palm nuts that a person needs a hammer for. It arrived with the bird, took no training, and cost nothing to carry. Every tool in this essay had to be built, learned, and kept charged.

The reach is already wider than conversation. Systems predict protein structures, find patterns in medical images, and help search spaces too large for manual inspection. These tools do not remove experiments or expertise. They alter where effort begins. A predicted structure can narrow a laboratory question. A generated prototype can make an idea concrete enough to criticize. Code can be drafted before every dependency has been understood. A scientist can spend less time locating one candidate and more time deciding which uncertainty deserves an experiment.

FIELD NOTES

A blank text field waits beneath a row of familiar icons.

The first answer arrives one line at a time, then stops in the middle of a sentence.

A generated photograph gets the light right and the fingers wrong.

Two versions of the same program sit open beside a list of changes.

before it settles





A layer that used to be the stable one.

This is where the privilege becomes strange. Most people who live after a major transition see only the stable layer. They cannot remember when the switch failed to mean light or when a recorded voice seemed uncanny. A generation living through machine intelligence can still see the joins. It remembers the first convincing output and the first obvious fabrication. It can compare work made before the tool with work made beside it. The comparison will not remain clean. New habits will become automatic. Children will meet capabilities as defaults that older adults experienced as ruptures. For a short period, both conditions exist in the same room.

Professions change inside that overlap. A designer can produce more variations before a meeting. A programmer can begin with a proposed function instead of an empty file. Review becomes a larger part of the craft. Faster production can create room for judgment, or it can become the reason more production is demanded. The person who once made every first draft becomes responsible for detecting the subtle failure that an effortless draft conceals.

AlphaFold's public database provides over 200 million predicted protein structures. Experimental structures still matter, but researchers gained a computational starting point at a scale no laboratory could produce one protein at a time.

NOTICE THIS

Save one result that would have seemed impossible ten years ago. Record the exact part that still required judgment. Keep both pieces together.

The costs are also being built into the ordinary layer. Large systems require concentrated computing power and vast collections of data. Their errors can spread at the speed of copying. Synthetic evidence can be made cheaply, while trust remains slow to repair. A useful assistant can become an authority it has not earned. These are not objections from outside the revolution. They are part of its design. Every convenience establishes dependencies. Every interface decides what effort becomes invisible. The unsettled period matters because habits, markets, and laws are still responding to one another. Later, many of these choices will look natural because they will be old.

The paragraph on the screen is selected and revised. A specific noun replaces a vague one. The unsupported claim is removed. One sentence remains exactly as the machine produced it because it does the work cleanly. The final page contains no reliable way to separate the origins. It has become a joint artifact, though responsibility still rests with the person who sends it. This small act is not the whole revolution. It is one of the places where the revolution becomes real. The future enters through repeated decisions about what to accept, what to question, and what should never be made effortless.

It is the brief chance to watch the new machinery become ordinary while its choices can still be seen.



The image is a full-frame, close-up photograph of a metal surface that has undergone significant oxidation. The surface is covered in a complex pattern of colors and textures. Dominant colors include deep teal and turquoise, which are interspersed with patches of vibrant orange and reddish-brown, characteristic of rust. The lighting is dramatic, coming from a low angle (raking light), which creates strong highlights and deep shadows, emphasizing the rough, pitted, and uneven texture of the corrosion. The overall effect is one of decay and the passage of time.

Oxidized metal · rust bloom and verdigris · raking light

Time, doing its work on something
that was supposed to last.



The Last People Who Remember Waiting

A vanished default survives for a while in the people who once expected it.

A MAN SITS FOR A PORTRAIT IN A STUDIO WITH A PAINTED BACKDROP BEHIND HIM. THE PHOTOGRAPHER CHOOSES THE MOMENT, AND NOBODY IN THE ROOM SEES THE RESULT. THE FILM LEAVES IN AN ENVELOPE AND COMES BACK WEEKS LATER AS A SET OF PRINTS IN SEVERAL SIZES. THERE IS ONE EXPRESSION, AND IT BECOMES THE WAY THIS FACE LOOKS TO EVERYONE WHO IS GIVEN A COPY. IT CANNOT BE CHECKED ON THE DRIVE HOME. IT CANNOT BE DELETED, FILTERED, OR QUIETLY REPLACED BY A BETTER FRAME FROM THE SAME AFTERNOON, BECAUSE THERE IS NO BETTER FRAME AND NO SAME AFTERNOON TO RETURN TO. COPIES GO TO RELATIVES WHO PUT THEM IN DRAWERS. THE PORTRAIT BELONGED TO A WORLD IN WHICH AN IMAGE COULD BE COMPLETE, PORTABLE, AND STILL UNABLE TO BE REVISED.

There was no single day when that world ended. The before-time arrived and disappeared unevenly. A household could have a computer without an internet connection, then a slow connection that occupied the telephone line, then broadband in one room. A mobile phone might be carried for emergencies and kept switched off to preserve the battery. Some people lived online while their neighbors still

mailed checks and waited for photographs to be developed. The boundary was never global. Access depended on income, infrastructure, age, and place. Even now, the network does not reach everyone equally. The phrase names a change in expectation, not a clean historical border. It marks the period before constant access became the assumed condition of ordinary life.



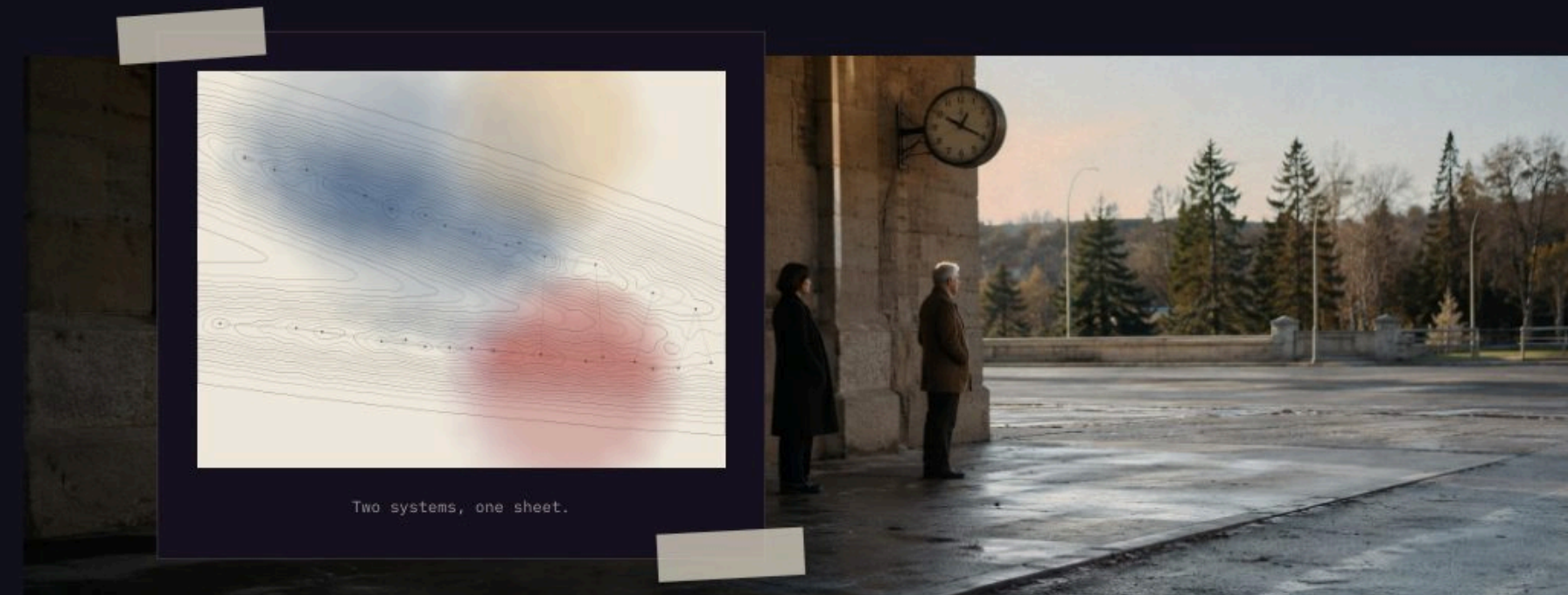
To leave home once meant becoming difficult to reach. A telephone rang in a particular building. If nobody answered, the caller tried later or left a message on a machine. Plans had to survive without continuous repair. A meeting place and time were agreed upon before anyone departed. Someone who ran late could not make lateness visible from the road. The others

waited or left. This required more patience than the present system and caused more needless worry. It also created intervals when a person's attention was not open to distant requests. Absence was not a setting selected from a menu. It was what happened when a body moved beyond the length of a wire.

Pew Research Center surveys found that 52 percent of U.S. adults used the internet in 2000. By 2025 the figure was 96 percent, though survey methods changed during that span.

Media carried the same delay. A roll of film held perhaps twenty-four or thirty-six exposures. The image remained latent until the roll was finished, processed, and printed. By the time a blurred frame returned, the room had changed and the people had gone home. Music was arranged on physical sides. A missed broadcast might be missed completely. A library search

ended where the shelves and catalog ended. Scarcity could be frustrating, especially when the needed fact was far away. It also gave each choice a visible cost. One photograph used one frame. One song occupied part of a tape. A copy required time, material, and a machine close enough to make it.



The network reduced those frictions in ways that are difficult to separate from daily competence. Directions adjust while the car moves. A family can see a child across an ocean. A medical result arrives without another trip to the clinic. Lost facts can be retrieved before an argument finishes. These are not small gains, and remembering the earlier world does not make it

better. Nostalgia edits inconvenience with the same confidence that a phone edits a photograph. It removes the missed connection, the inaccessible record, and the person stranded without help. The before-time contained privacy and isolation in the same silence. The present contains connection and surveillance in the same device.

Home broadband was reported by 1 percent of U.S. adults in March 2000. Smartphone ownership rose from 35 percent in Pew's first 2011 survey to 91 percent in 2025.

Figure 5. Things that did one thing each. Every function here now lives in a single object in a pocket.



SPECIMEN Bubo bubo
COMMON Eurasian eagle-owl
RECORDED July 20, 2014, 1:31
p.m., captive

The tufts are not ears. The ears are under the feathers, set at different heights on each side, so a sound arrives at one a fraction before the other and the difference is a position. The bird is holding a map of a room it has not looked at.

What changed most was not the number of screens. It was the default answer to uncertainty. A forgotten name once remained forgotten until someone remembered it. A disagreement could end with no verdict. The location of a friend was unknown unless the friend reported it. Now the first movement is often toward a search field, a map, or a message. The impulse arrives before the sentence has finished forming. This does not mean memory has failed. It means retrieval has become part of thought. The older condition survives as a bodily hesitation in people who remember when asking the network was not available, not quick enough, or not yet the obvious thing to do.

FIELD NOTES

A road atlas rests open across the passenger seat with one fold split white.

The telephone rings in an empty kitchen until the answering machine starts.

A photograph returns from processing with a thumb across one corner.

Three people wait beneath a clock because none can report that the bus is late.

before always online





Returned, years later, by the archive that kept it.

Children who grew up before constant connection also grew up before ordinary life produced a continuous public record. Embarrassing remarks could be repeated, but they were less likely to remain searchable. A photograph taken at a party might exist in one envelope. School friendships ended when families moved and addresses were lost. This made some harm easier to escape and some people easier to disappear from. Memory belonged more heavily to those who were present. Accounts conflicted. Details changed. The record was thin and human. A later generation inherits a denser archive, but not a neutral one. Platforms decide what remains visible. Cameras appear at uneven moments. What survives is abundant without being complete.

The last generation to remember both conditions is not defined by a birth year. It is a moving group whose members crossed the boundary at different ages. They learned to wait beside a landline, then learned to carry a device that made waiting optional. They memorized routes and wrote letters before navigation and messages became continuous services. This gives them no special wisdom. It gives them comparison. They can feel when a convenience replaces a skill because both actions remain available in memory. They can also remember how quickly an awkward tool became a social requirement, then became almost invisible.

The first SMS message was sent from a computer on December 3, 1992. Early text messages carried a 160-character limit, a technical boundary that helped shape a new compressed style of writing.

NOTICE THIS

Find one object whose instructions assume no network connection. Follow the sequence exactly as printed. Notice which delays and decisions the object expects a person to manage.

That memory will thin. Objects can be preserved, but the expectation around them is harder to display. A rotary telephone in a museum does not reproduce the experience of waiting beside it. A paper map does not restore a world in which nobody in the car could see the road ahead. The important artifact is the former rule. Calls happened at places. Photographs arrived later. Plans contained silence. Once the people who lived under those rules are gone, the objects will remain as evidence without witnesses who remember their weight. The before-time will become a style that can be imitated more easily than a condition that can be understood.

The car reaches the next town without using the atlas. A phone has been giving directions from the cup holder, its voice arriving a few seconds before each turn. The paper map remains open anyway. Its route and the blue line agree for most of the distance. At one junction they divide. The driver follows the phone and passes a field that the old road never reached. Nothing dramatic marks the change. One system continues to work while another becomes unnecessary. The before-time ends this way in memory too, not with a closing date, but with an old instruction still legible beside a habit that no longer needs it.

For a little while, one generation can still describe both worlds from inside them.

Absence was not a setting selected from a menu. It was what happened when a body moved beyond the length of a wire.



FIELD NOTE

A STREET WITH NOTHING ON IT · OCTOBER

stopped for no reason

The walk had been an errand until the person beside me stopped. I took two more steps before the leash and the silence made me turn. There was no car coming and no one calling from a yard. We stood under the utility wires while a few dry leaves moved along the curb. Later, I could not remember what we had gone out to get. I could still hear the faint electrical hum above us and see the empty pavement ahead. The two figures stayed in the same place until the leaves reached the storm drain.



— INTEGRATE WHAT IS LEFT

Part IV

While We're Here

Nothing here is permanent.
Nothing here is therefore smaller.

Distance, arrival, ordinary rooms, and the time we get to spend
in them together.

01 The Body Cannot Skip the Hill

02 While We're Here



The Body Cannot Skip the Hill

A destination changes when the body has to carry time all the way to it.

A PUBLIC STAIR DROPS AWAY BETWEEN TWO HANDRAILS. THE SHADOW OF THE PERSON AT THE TOP LIES DOWN IT, ACROSS HALF THE FLIGHT. MOSS HAS COME IN AT BOTH EDGES AND STOPPED WHERE THE FEET BEGIN. EACH PERSON ARRIVED AT THE SAME LANDING, TOOK THE SAME FIRST STEP, AND WENT DOWN. NO SINGLE CROSSING CHANGED THE STAIR ENOUGH TO SEE. THE ROUTE IS VISIBLE AS THE STRIP THAT STAYED CLEAR. PILGRIMAGE BEGINS WITH THIS PLAIN FACT. NOTHING REQUIRES IT ANYMORE — WHOLE SYSTEMS EXIST SO THAT NOBODY HAS TO WALK ANYWHERE — AND THE STRIP KEEPS ITS SHAPE ANYWAY. A PLACE IS REACHED BY PASSING THROUGH THE DISTANCE BEFORE IT. THE BODY CANNOT SKIP THE HILL BECAUSE THE DESTINATION MATTERS. IT CANNOT ARRIVE EARLY BY UNDERSTANDING THE MAP. SHOES WEAR. WATER IS CARRIED. WEATHER REACHES THE SKIN. AN INTENTION THAT BEGAN AS A SENTENCE IS GIVEN WEIGHT, DURATION, AND A SERIES OF SURFACES.

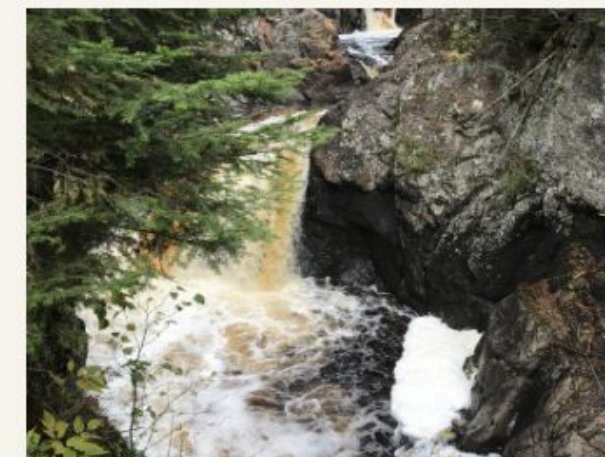
Travel usually treats distance as a problem to reduce. Faster routes are preferred. Waiting is measured as lost time. The destination is the useful part, and the miles between are charged against it. Pilgrimage reverses that arrangement. The route is not an obstacle placed before the meaningful place. It is the method by which the place acquires meaning. A

shrine reached after forty days is physically the same structure that could be reached by bus. The arrival is different because the traveler has spent forty mornings approaching it. The path has occupied meals, sleep, and the condition of the feet. Time has not been removed from the journey. It has been made into one of its materials.



Traditional pilgrimages give the movement a form. The Shikoku Henro circles an island through eighty-eight temples. The Camino de Santiago gathers many routes toward one city. Markers reduce the number of decisions. Stamps, repeated prayers, shared clothing, and named stages make progress countable. These rules do not make every pilgrim's purpose the same. One person walks from religious obligation. Another walks after a

death, a recovery, or a change that has not found words. The route can hold different reasons because it asks for the same next action. Wake, then walk to the next place. A numbered sequence lends the private reason a public shape, making a month of effort possible to describe without explaining the reason itself. The larger question is carried without being solved at every step.



Worn by something that only went one way.

The Shikoku Henro links 88 temples across Japan's four Shikoku prefectures. The full circuit covers approximately 1,200 kilometers and generally takes more than 40 days to complete on foot.



**Distance gives
an intention
enough time to
become physical.**

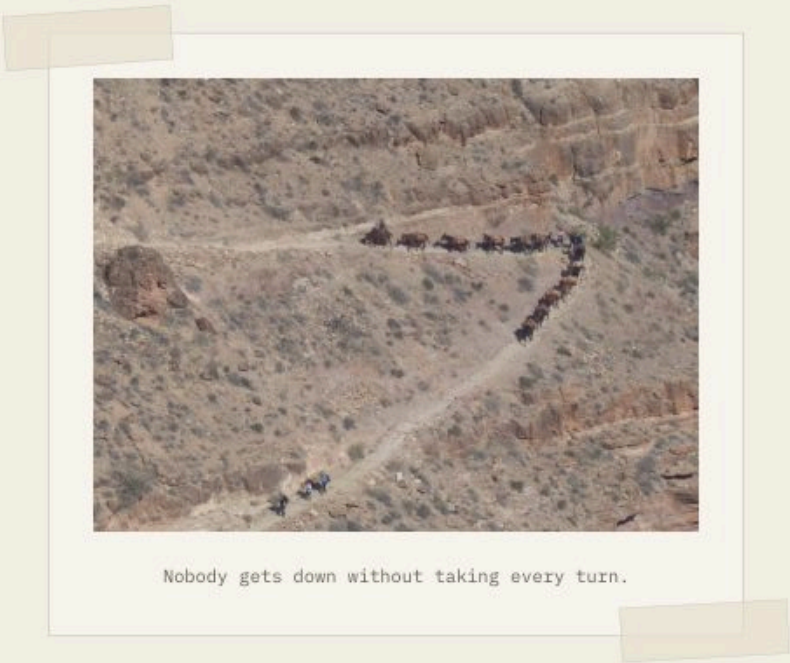
The body supplies a measure that cannot be argued with for long. A blister changes pace. Heat decides when to stop. Hunger makes the next town more important than the distant end. This narrowing is part of the usefulness. Ordinary life allows a difficult thought to be interrupted by a dozen smaller claims. On a long

route, attention returns to water, grade, weather, and the next marker. The concern that began the journey remains present, but it loses the ability to occupy every minute. It has to travel beside practical work. By evening, the walker has not escaped the question. The question has been placed among other facts.



Pilgrimage is also built by strangers. Historic routes required wells, bridges, hospitals, and places to sleep. Towns learned the rhythm of arriving feet. Food was prepared for people who would leave the next morning. A route becomes durable when hospitality is repeated along it. The walker depends on people who did not begin the journey and may not share its reason. This

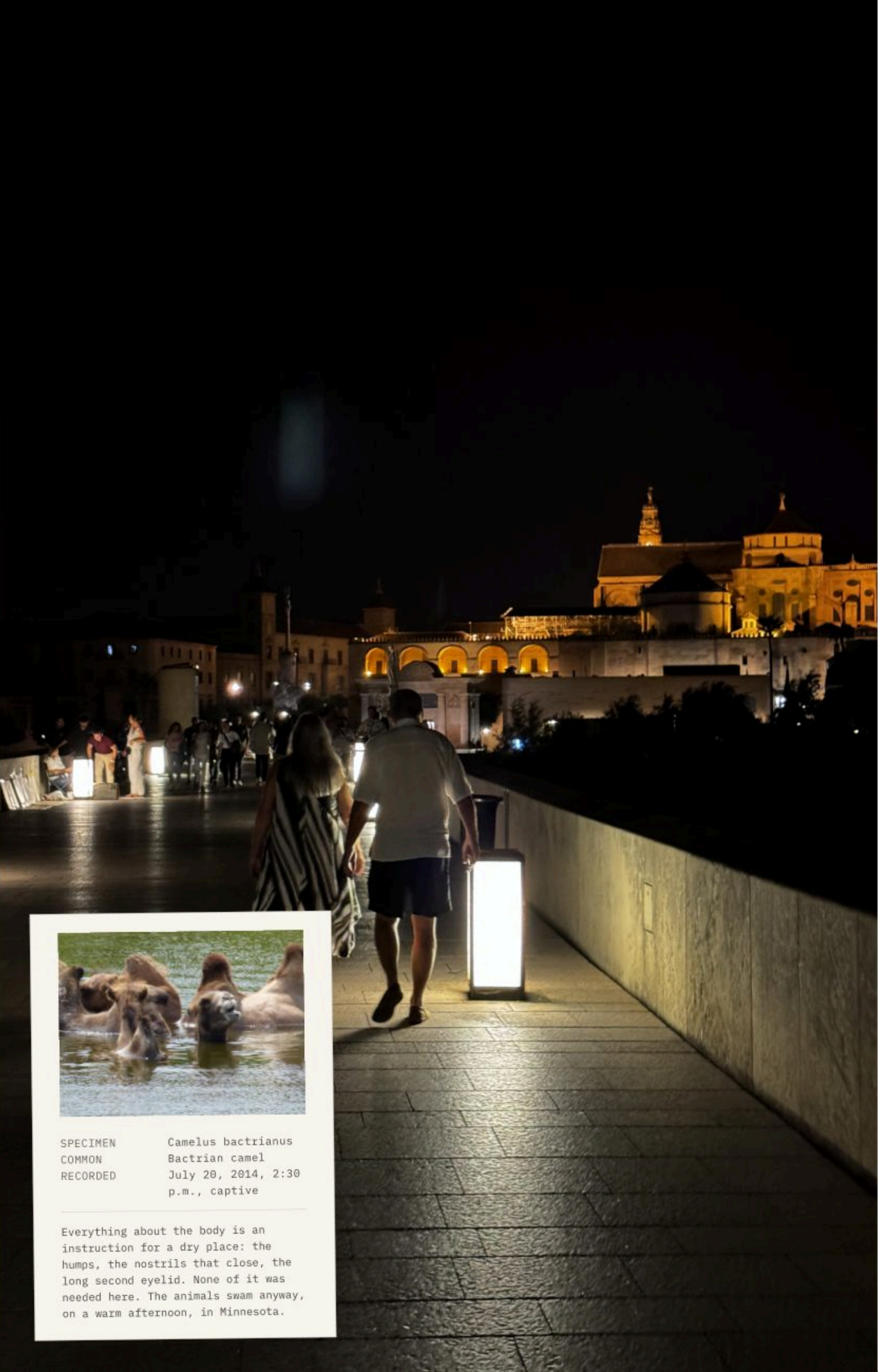
dependence is easy to overlook in accounts centered on private transformation. The path is a social structure. Someone maintains the marker. Someone recognizes the tired person at the door. Solitude on a pilgrimage is supported by a long chain of other people's work.



Nobody gets down without taking every turn.

Spain's official tourism site describes roughly 800 kilometers between the Pyrenean starting routes and Santiago de Compostela. The Compostela requires stated spiritual or religious motivation and at least the final 100 continuous kilometers on foot.

Figure 6. Crossing the Roman bridge at Córdoba, after dark. The route asks everyone for the same next step.



SPECIMEN Camelus bactrianus
COMMON Bactrian camel
RECORDED July 20, 2014, 2:30
 p.m., captive

Everything about the body is an instruction for a dry place: the humps, the nostrils that close, the long second eyelid. None of it was needed here. The animals swam anyway, on a warm afternoon, in Minnesota.

The shared route changes encounters between walkers. Names may be exchanged after the destination and the day’s distance are already known. Status is difficult to carry in the same way when clothing is dusty and everyone is looking for the next bed. People separate, meet again, and continue without promising friendship. The route provides enough common structure for brief trust. One person points toward water. Another waits at a confusing turn. Advice passes backward from those who crossed the hill earlier. These exchanges are small, and most leave no record. The path keeps producing them because different lives have been given the same temporary direction.

FIELD NOTES

- A boot rests on a stone step polished lower at its center.
- Two walkers compare stamps in folded paper booklets beside a fountain.
- A shrine is set into the rock at one of the turns, where anyone walking would stop anyway.
- The destination remains hidden while the path continues through rain.

earned by distance





Come a long way, to stand on old stones.

Religious pilgrimage gives this direction a sacred destination, but the human pattern appears elsewhere. People walk to a childhood home before it is sold. They travel to a battlefield, a grave, or the coast where a migration began. A runner returns to the same long race each year. Fans cross a country to stand in a stadium associated with a person they never met. The destinations differ in seriousness, but the structure is recognizable. A place has been asked to hold something that ordinary language cannot carry alone. Effort creates a boundary around the visit. The person arrives having paid in time rather than simply having appeared.

This can be manipulated. Commercial routes can sell difficulty as identity. A crowded path can become a queue of private performances. Suffering can be treated as proof that the traveler deserves an answer. The road owes none. Distance does not make a person wise, and pain is not automatically meaningful. The value lies in the conditions the route creates. Progress becomes slow enough to notice. Dependence becomes visible. A purpose has to survive boredom and bad weather. The traveler can still return unchanged. Even then, the miles were real. The claim made at the beginning was tested against the body that carried it.

UNESCO's French Santiago property includes 71 selected monuments and seven sections of the Chemin du Puy totaling nearly 160 kilometers. Churches, hospitals, and bridges record the practical infrastructure built around walking pilgrims.

NOTICE THIS

Choose one familiar destination and reach it by a slower route. Carry what is needed and do not shorten the last section. Let arrival occur only after the distance has been crossed.

Arrival needs a fixed edge. Without one, a long walk can dissolve into mileage. The temple gate, cathedral square, summit cairn, or plain stone gives the route somewhere to stop. People touch the surface, sit down, remove a pack, or stand without knowing what action belongs next. The destination often looks smaller than the anticipation surrounding it. This is useful too. The visible place cannot contain every reason brought toward it. It only marks that the distance has been completed. The walker is left with tired legs, a changed calendar, and the difference between the person who departed and the person now standing there.

The yellow arrow points through a residential street where laundry moves above parked cars. Nothing in the lane announces that it belongs to an ancient route. A walker follows it past a bakery and a road crew lifting stones. Pilgrimage does not remove ordinary life from the path. It threads purpose through it. The sacred route uses the same pavement as errands and deliveries. That may be why it can return a person to daily life without pretending daily life has been replaced. The destination matters, but the road is made from the world already here. The final marker is reached one ordinary step after another.

**Tomorrow the body will turn back
toward ordinary rooms, carrying the
route in its feet.**





While We're Here

The world does not become ordinary when it is understood. It becomes more fully present.

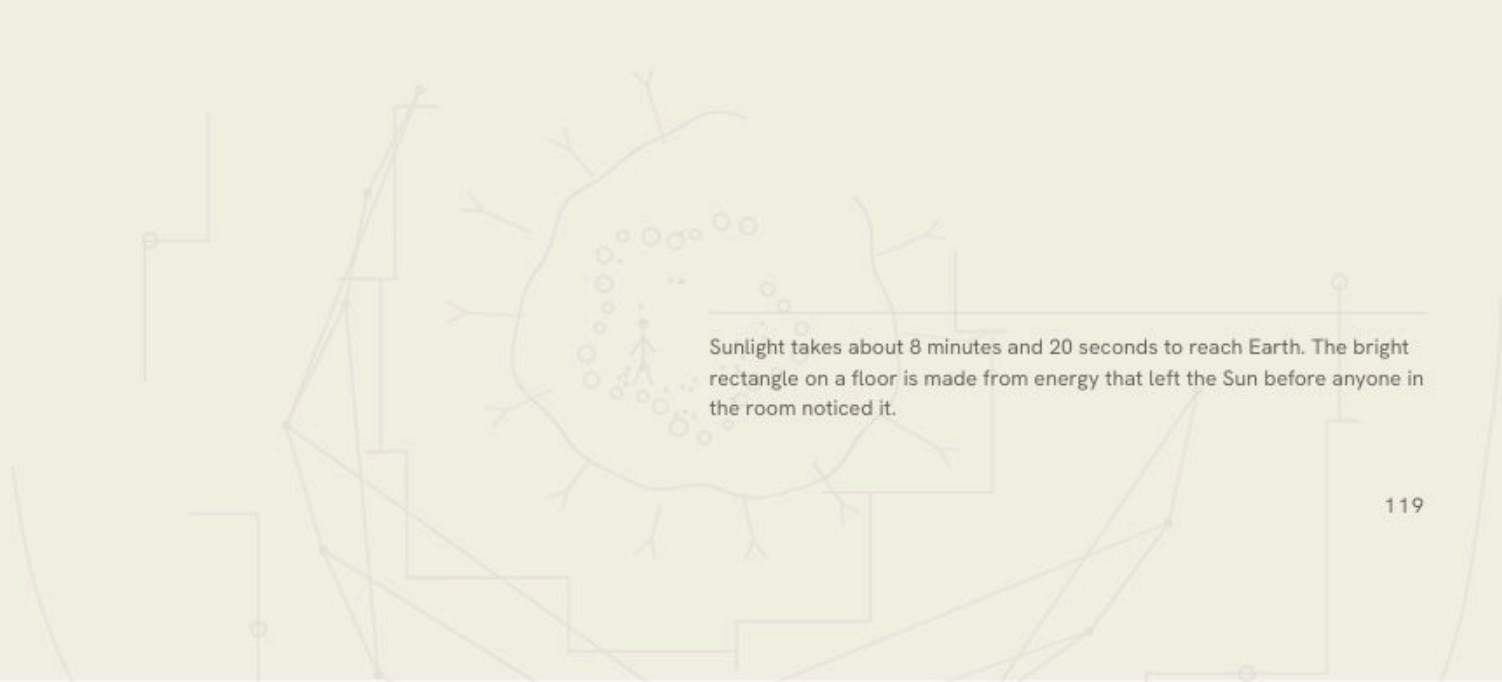
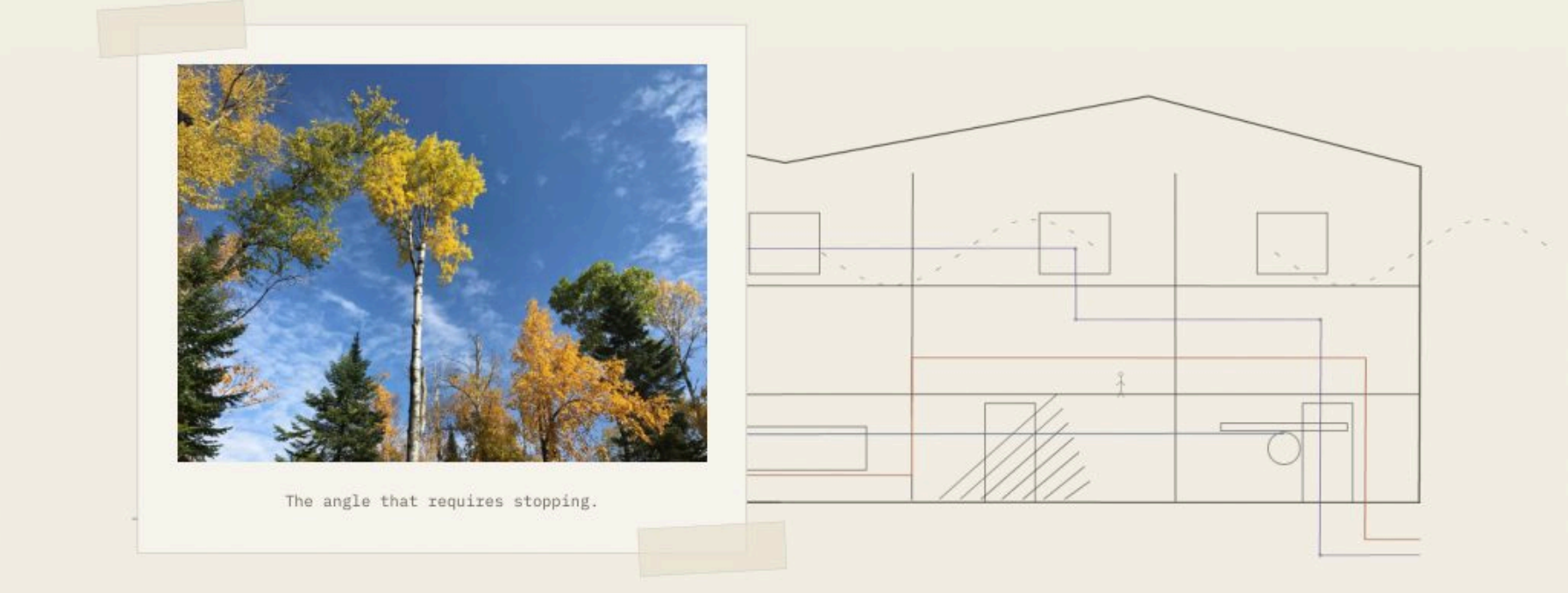
A DOG LIES IN A RECTANGLE OF AFTERNOON LIGHT ON THE FLOOR. THE BRIGHT EDGE MOVES SLOWLY ACROSS THE BOARDS. WHEN IT REACHES HER BACK, SHE STANDS, TURNS ONCE, AND SETTLES INSIDE THE WARMER PART. NOTHING ELSE IN THE ROOM APPEARS TO CHANGE. THE LIGHT HAS TRAVELED FROM THE SUN FOR ABOUT EIGHT MINUTES. THE FLOOR IS ROTATING WITH EARTH, AND THE HOUSE IS MOVING AROUND THE SUN. AIR ENTERS THROUGH A VENT. WATER WAITS UNDER PRESSURE INSIDE THE WALL. A SIGNAL PASSES THROUGH A CABLE BENEATH THE STREET. THE STILL ROOM IS A MEETING PLACE FOR MOTIONS TOO LARGE, SMALL, OR FAMILIAR TO HOLD ATTENTION FOR LONG. THE DOG FINDS THE USEFUL PART AND CLOSES HER EYES.

Looking closely does not make the ordinary world fragile. It makes the word ordinary less precise. A glass of water is plumbing, weather, geology, treatment, and a hand lifting it from the table. A tree beside the sidewalk is sunlight held in wood, roots working through compacted soil, insects using the bark, and years recorded without numbers. None of these

descriptions replaces the glass or the tree. They add depth to what remains directly in front of us. Explanation is not the end of wonder, and wonder is not a claim that causes must stay hidden. The visible thing can be understood as a surface where many processes have arrived at once.

Attention cannot keep all of those processes present. It should not. A nervous system that treated every hum, shadow, and pressure change as new would be unable to finish breakfast. Familiarity lowers the volume so action can continue. The cost is that whole systems disappear while they are working well. Electricity becomes a switch. Clean water becomes a faucet. A

loved person becomes the sound of movement in another room. Their reliability lets them recede. Then a failure, an absence, or a deliberate pause makes the supporting structure visible again. The aim is not permanent intensity. It is the ability to let one familiar thing return to the foreground before loss is the only force strong enough to bring it back.



Human life occupies a narrow bracket. Before it, the lake froze and opened without us. Roads were laid along older paths. Languages arrived through mouths that are gone. After it, the same systems will continue in altered forms. This scale can make one life look negligible, but size and importance are not the same measure-

ment. A small interval can contain every person a person knows, every room remembered, and every choice that changed the next day. The bracket is not an error in the larger timeline. It is the only section available for experience. Its limits do not make its contents less real. They give the contents a location.

Technology changes what can happen inside that interval. A machine can retrieve a record, extend a voice across distance, or produce an image that never passed through a camera. The change is large and still enters through ordinary gestures. Someone types and checks before deciding whether a result belongs. New systems will become furniture as earlier ones did. Their

presence does not cancel the older work of being alive with other people. Messages still arrive too late. A photograph still matters because of who is in it. More knowledge does not remove the need to choose where attention goes. The expanded world remains limited by the hours of one day.



Thirty minutes, on a Thursday afternoon.

Earth turns once relative to the Sun in 24 hours and travels around the Sun at roughly 29 to 30 kilometers per second. Stillness at human scale occurs inside continuous planetary motion.

Figure 7. The far wall, for under a minute. The room is doing this whether or not anyone is in it.



SPECIMEN	Cynomys ludovicianus
COMMON	Black-tailed prairie dog
RECORDED	July 20, 2014, 2:44 p.m., captive

The colony posts sentries, keeps its burrows, and calls with different sounds for different threats. None of that stops for this one, flat on warm sand with its legs out behind it, doing nothing whatever on an ordinary afternoon.

Most of those hours are not milestones. They are preparation, maintenance, repetition, and return. Food is put away. The dog is walked. A parent calls with an update that nothing is wrong. Work is revised after the meeting. A person stands at the sink while rain moves across the yard. These intervals are easy to treat as the space between the important parts. In practice, they are the material from which a life receives its texture. The remembered events are held in place by hundreds of days that did not announce themselves. A relationship is built as much by the repeated greeting as by the photograph chosen to represent it.

FIELD NOTES

A dog moves from one patch of sunlight to the next as the floor cools.

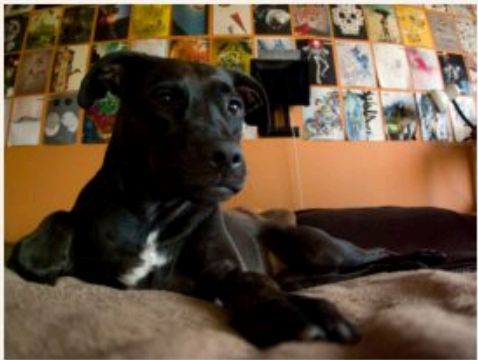
Water strikes the window, joins another drop, and changes direction.

Someone in the next room closes a cupboard and begins humming.

The evening light reaches the far wall for less than a minute.

still in progress





Resting, and not in charge of most of it.

The hidden systems inside the body continue through all of this. The heart contracts before a sentence and after it. Breathing adjusts to stairs, sleep, laughter, and fear. Cells repair tissue without waiting for a plan from consciousness. The body is not a silent vehicle carrying a separate observer. It is an active population keeping the observer possible. Its processes are reliable until they are not, and no amount of attention can place them fully under control. Care begins with that limit. A body can be fed, rested, exercised, examined, and still remain partly beyond prediction. Being alive includes depending on work that cannot be personally supervised.

Other people are beyond supervision too. They carry private rooms of memory, worry, and attention that can be approached but never occupied. Love does not remove that distance. It makes the distance consequential. A person can be known through habits, stories, expressions, and years of shared decisions, while still remaining more than the version held in another mind. This is why presence matters in such ordinary forms. Listening through the end of a sentence. Sitting nearby without improving the silence. These acts do not solve the temporary nature of a relationship. They are how the relationship exists while it is available.

A healthy resting adult typically breathes 12 to 20 times each minute, moving about 500 milliliters of air per breath. Most of this work continues without entering conscious attention.

NOTICE THIS

Stand where an ordinary room meets daylight. Stay until one visible condition changes. Do not name the lesson. Record only what moved.

No instruction can make a person notice everything in time. Attention wanders. Days are wasted. Familiar voices are answered impatiently. Beauty is missed because a notification arrives or because the mind is occupied with something real. The demand for perfect presence would turn noticing into another failure to manage. A smaller practice is enough. One thing returns. A patch of light is seen moving. The route around the lake feels different in wind. A machine's hum becomes audible after years in the room. The world does not need continuous appreciation to be real. It needs only occasional openings through which the ignored detail can be admitted again.

The light leaves the floor. The dog remains where it was, now resting in shadow. In the kitchen, water runs long enough to fill a pot. Someone outside calls to another person across the street. A plane passes above the roof without appearing between the branches. None of these events asks to be kept. Most will be gone before the room is dark. Their disappearance does not make them incomplete. They occurred inside the available bracket and altered what followed by a small amount. The pot will boil while the dog wakes. Someone will call a name from the next room. While we are here, we answer when we can.

For now, that is where we are.

A night photograph of a bonfire in a field. The fire is bright and intense, with a large pile of dark, charred wood or brush in the center. The fire's glow illuminates the surrounding tall grass. In the background, there are several buildings, some with lit windows, and a utility pole with wires. The sky is dark, and the overall scene is a mix of warm firelight and cool night tones.

**The temporary
thing is not less
real. It is the only
kind we have.**

the last warm night



I carried the chairs out before the light was gone and left my jacket inside. The air was cool enough to notice but not cold enough to send us back through the door. The fire took slowly. One log hissed at the cut end, and smoke moved toward the trees before lifting above them. We sat without moving the chairs closer. A screen door opened once behind us and closed again. By the time the flames settled below the rim, the boards of the deck had gone dark and the metal chair arms were cold.

HANDED OVER

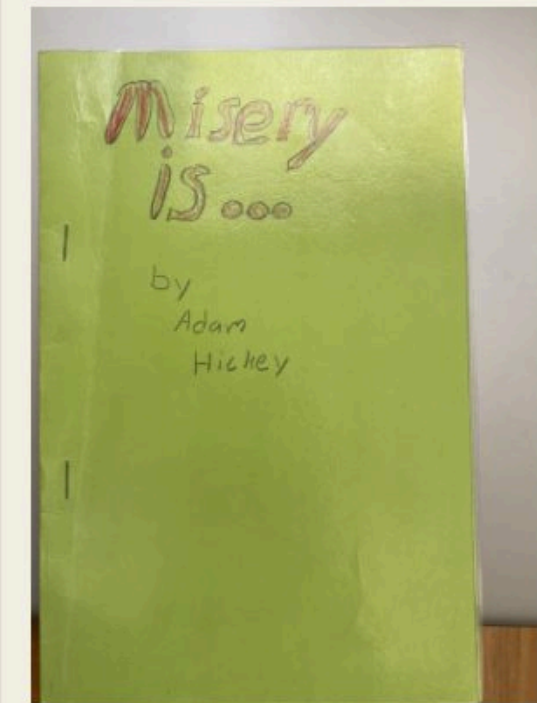


01

To the Love of My Life

Made by Fabiola. One copy, for one reader.

PHOTOGRAPHED 20 JUNE 2017, 2:39 P.M.



02

Misery Is...

Written and bound by the author as a child, and kept.
Inside, in pencil: "Misery is when your mom embarrass
younn front of your friends."

PHOTOGRAPHED 4 OCTOBER 2021, 1:56 P.M.



**You can walk around a familiar lake
believing almost nothing is happening,
while simultaneously moving through
thousands of invisible systems.**

Essays on attention, weather, water, machines that are learning to think, and the ordinary shared days that turn out to be the whole thing. A book for reading straight through, or for opening at random on a Tuesday.

LOOK CLOSER. STAY CURIOUS. BE KIND.